LFSUPC PHYSICS WORKING GROUP • MONDAY, AUGUST 24, 2026

π/K/p Spectra Analysis

$\sqrt{s_{NN}}$=200 GeV O+O Collisions for Space Radiation Studies

Andrew Liggett

AGENDA

What We'll Cover

1. **Physics motivation for O+O**
   Oxygen as a galactic cosmic ray proxy -- why O+O collisions matter for space radiation and shielding models.

2. **Detector, tracking, and event-quality checks**
   Acceptance, PID, dE/dx and BTOF fit diagnostics, and event QA.

3. **Yield extraction and corrected spectra**
   From raw TPC/BTOF yields through corrected mT-m0 spectra -- now complete for π/K/p.

4. **Status and next steps**
   Where the analysis stands -- eTOF calibration ongoing, corrected spectra underway.

Space Radiation Motivation

1. Oxygen as a galactic cosmic ray proxy

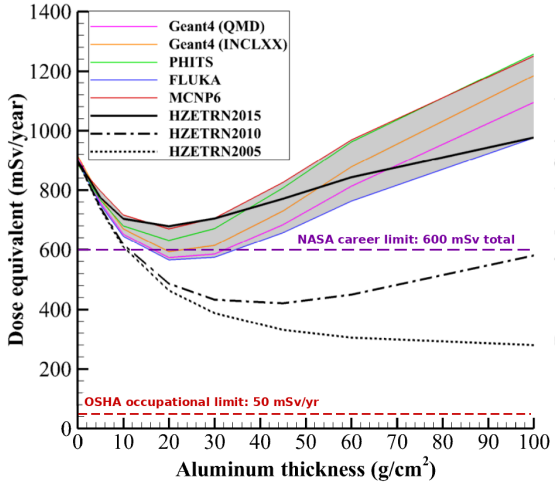
Oxygen is a major heavy-ion component of galactic cosmic rays -- O+O collisions probe the same nuclear fragmentation GCRs undergo in spacecraft shielding and human tissue.

2. Benchmarking models and shielding design

Identified π/K/p yields benchmark hadronic models (FLUKA, GEANT4, PHITS) and feed into shielding design and astronaut dose/ risk estimates for long-duration missions.

3. Quantifying the fractional dose

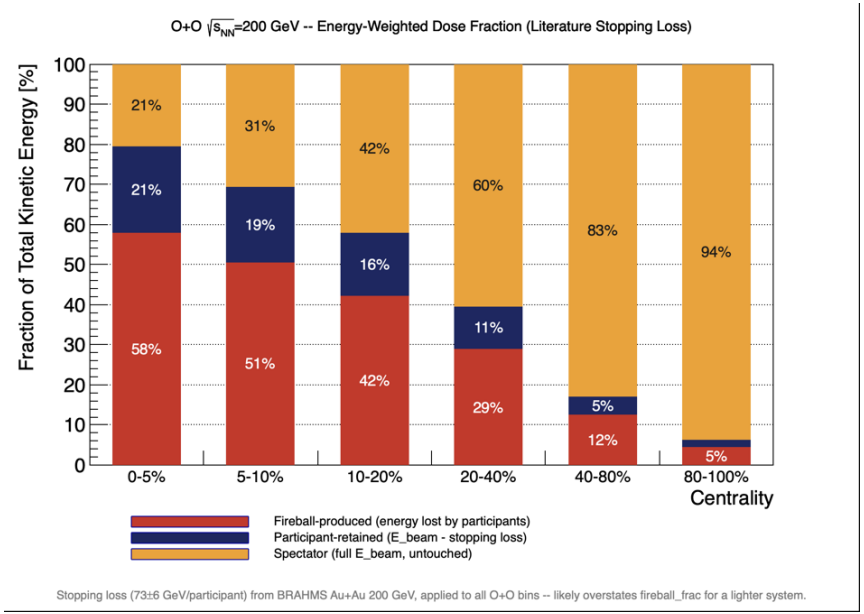
Rapidity stopping and pT spectra measure participants and fireball energy; spectators near beam rapidity set the forward dose.



(a)

Dose equivalent vs. aluminum shielding -- Monte Carlo codes (incl. FLUKA) vs. HZETRN versions. Norbury, Slaba et al. (2019); OSHA and NASA limit lines added (NASA's is a career total, not annual).

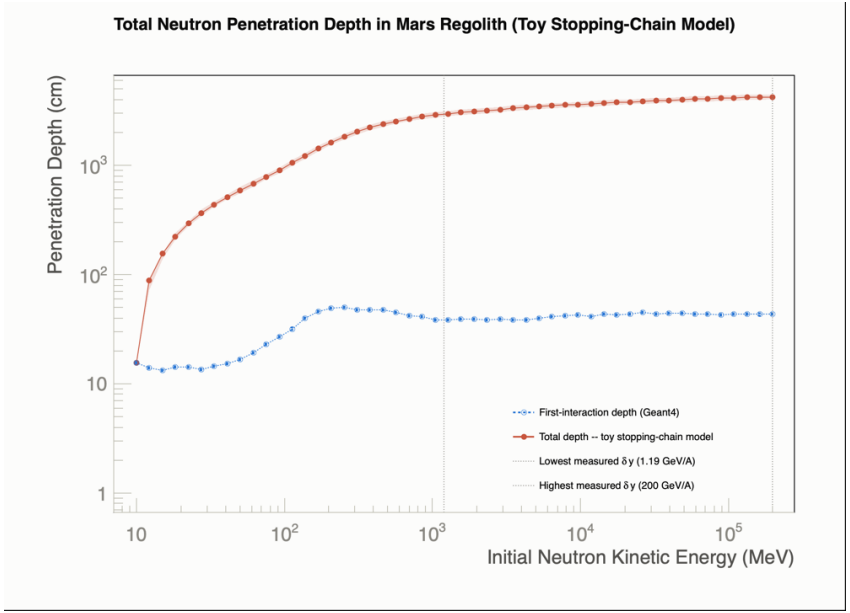
Fractional Energy Dose by Centrality



Energy-conservation split by centrality: spectators near beam rapidity keep full beam energy, setting the forward dose.

Literature nuclear-stopping fractions applied to a Glauber Npart/Nspec split

Neutron Penetration Depth in Mars Regolith



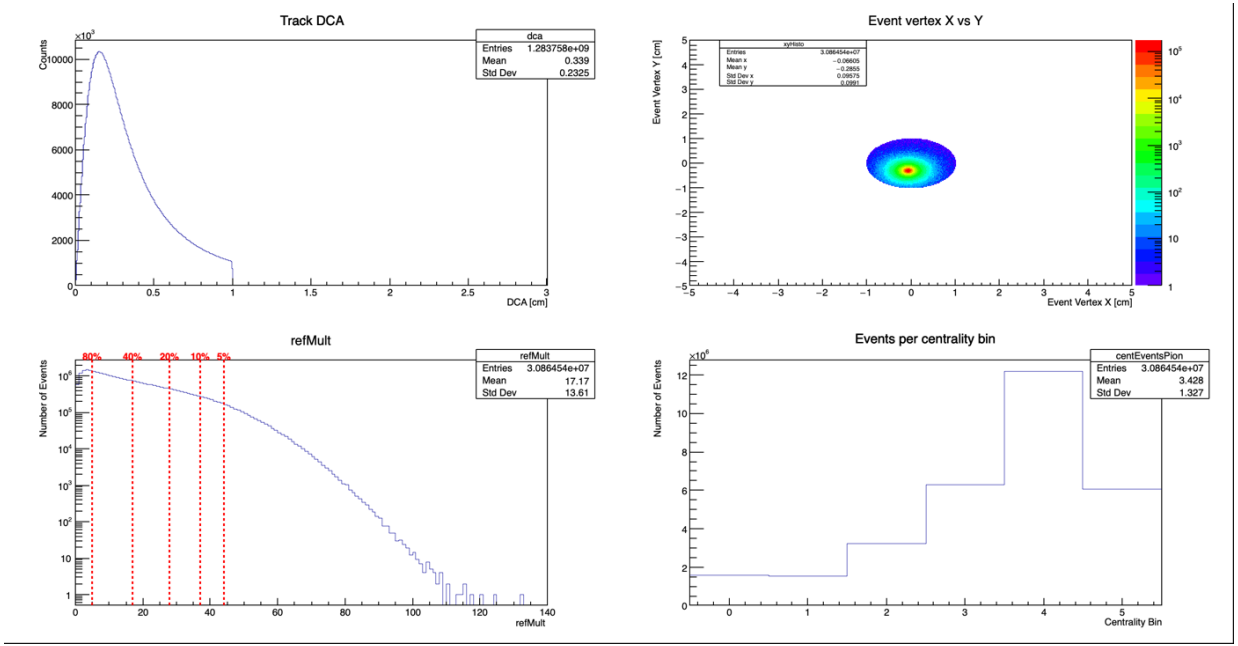
Toy stopping-chain model combining Geant4's first-interaction depth with literature baryon-stopping loss estimates total penetration depth.

Geant4 v11.3.2, FTFP_BERT_HP; δy anchors from BRAHMS, NA49, E917, Gao et al.

Event Selection Cuts

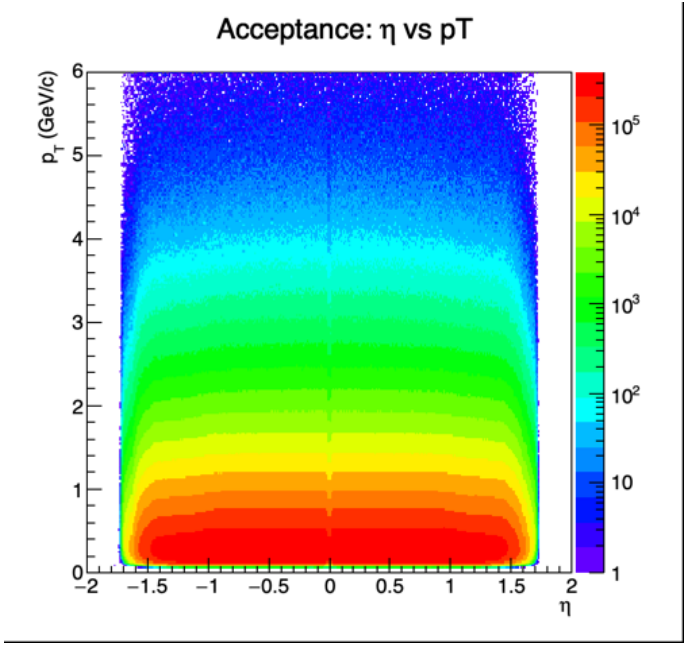
Cut	Value
Vertex Vz	[-2, 2] cm
Vertex Vr	<= 1.0 cm
Trigger ID	860003
Global DCA	<= 1.0 cm
Eta Range (TPC)	[-2.05, 2.05]
nHits Fit	>= 10
nHits dE/dx	>= 10
Fit Max Ratio	>= 0.52 (nHitsFit / nHitsPoss)
Centrality Bins	0-5%, 5-10%, 10-20%, 20-40%, 40-80%, 80-100% (raw refMult)

Event QA: Trigger Cut Verified



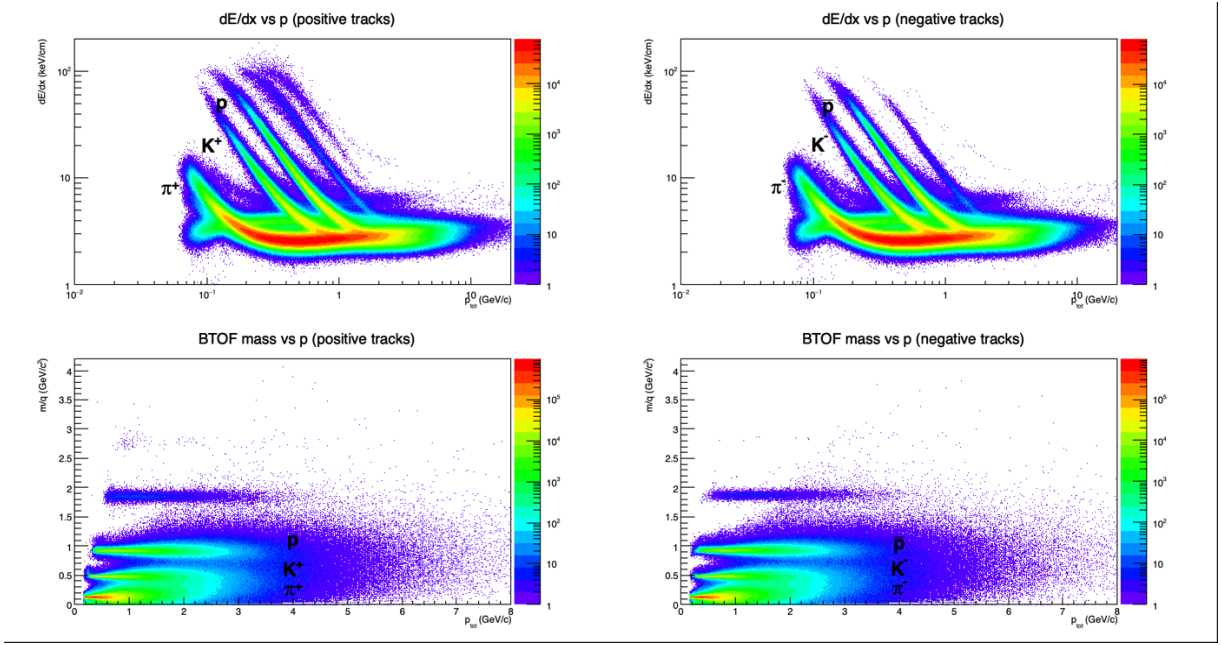
PION event QA: DCA, vertex XY, refMult centrality cuts, and events per centrality bin.

Acceptance: η vs pT



TPC-only acceptance in pseudorapidity (eta) vs pT -- defines the kinematic reach of the analysis.

Particle ID: dE/dx and BTOF Mass

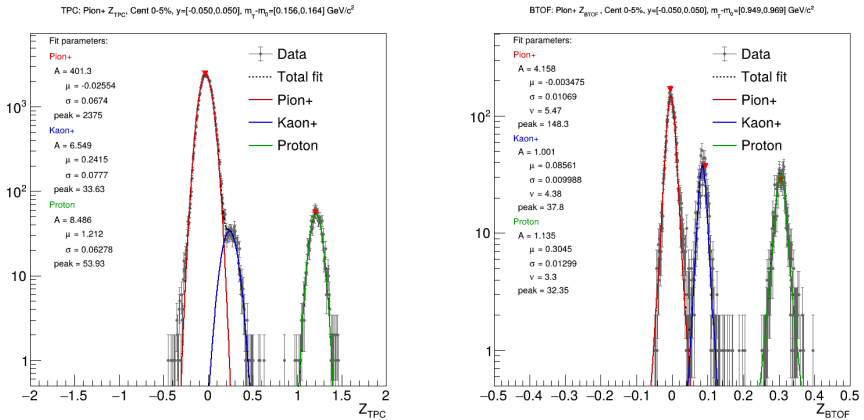


TPC-to-BTOF PID handoff momenta (pi/K/p): 2.71/2.0/4.0 GeV/c (+), 4.0/1.56/0.0 GeV/c (-) -- pi/K/p bands are cleanly separated.

Extracting Yields: dE/dx & Inverse-Beta Fits

$$Z_{\mathrm{TPC}} = \ln\left[\frac{(dE/dx)_{\mathrm{meas}}}{(dE/dx)_{\mathrm{Bichsel}}(p,\ \mathrm{species})}\right]$$

$$Z_{\mathrm{BTOF}} = \left(\frac{1}{\beta}\right)_{\mathrm{meas}} - \left(\frac{1}{\beta}\right)_{\mathrm{expected}}(p,\ \mathrm{mass}) \qquad \text{where } \left(\frac{1}{\beta}\right)_{\mathrm{expected}}(p, m) = \sqrt{1 + \frac{m^2}{p^2}}$$

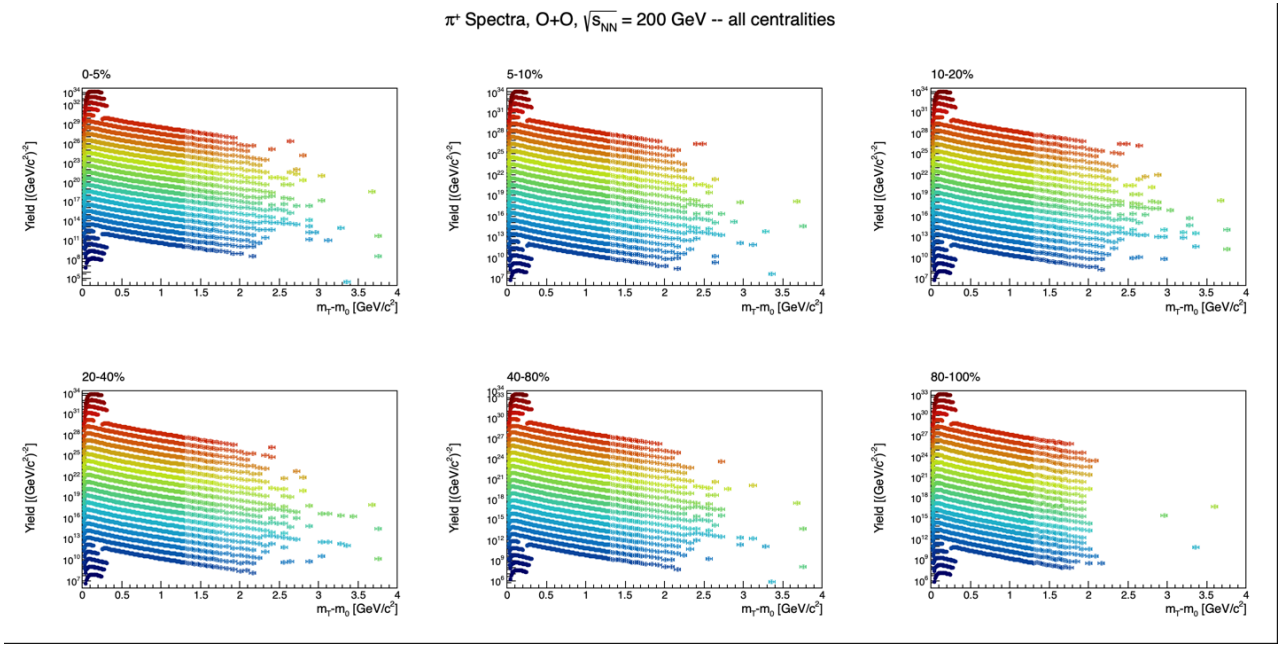


Gaussian (TPC) and Student's-t (BTOF) fits, pion+/kaon+/proton, 0-5% centrality. TPC pi+K fit range: p <= 0.75 GeV/c.

Spectra Corrections: Status

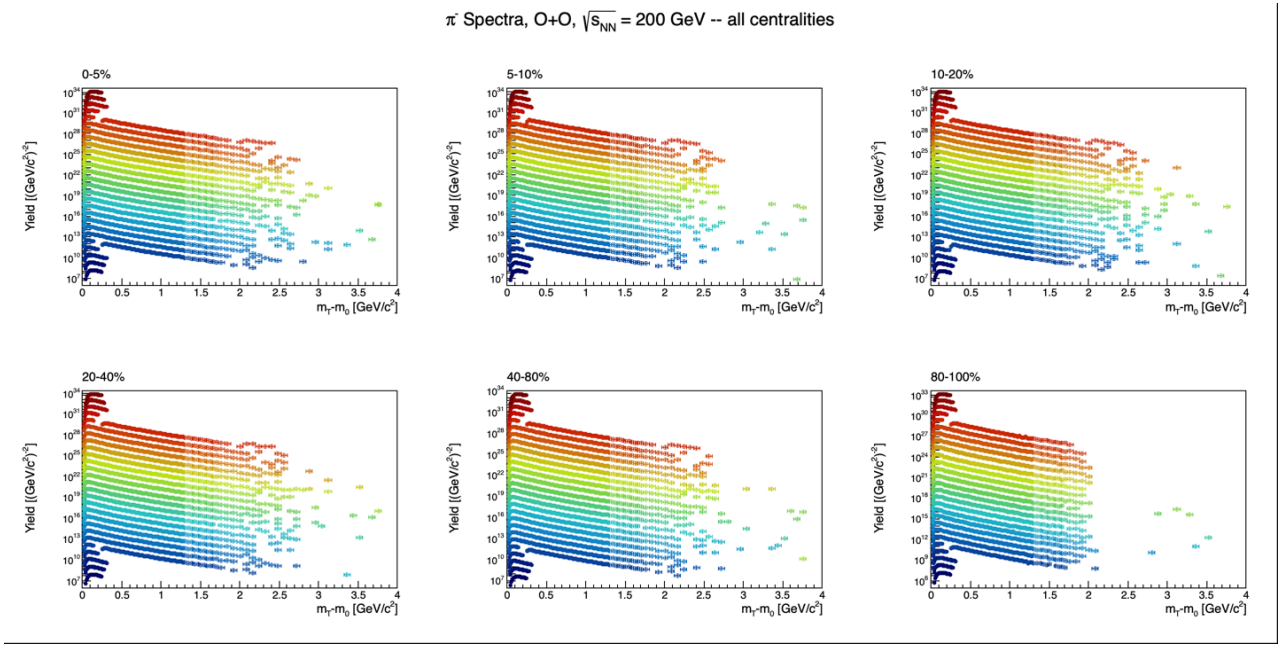
Correction	Status	Notes
BTOF efficiency	**Done**	Data-driven: ratio of TPC-identified tracks with a matching BTOF hit to all TPC-identified tracks, fit vs mT-m0.
TPC efficiency	Not yet applied	Will use the embedding already produced in P23ic (library SL23c) to derive the efficiency correction.
Energy loss	Not yet applied	Uses the same P23ic/SL23c embedding as TPC efficiency -- applied in the same call.
Hybrid feed-down	Not yet applied	Also draws on the same P23ic/SL23c embedding; will be enabled once the correction is applied.

Pion+: Combined All-Centralities Spectra View



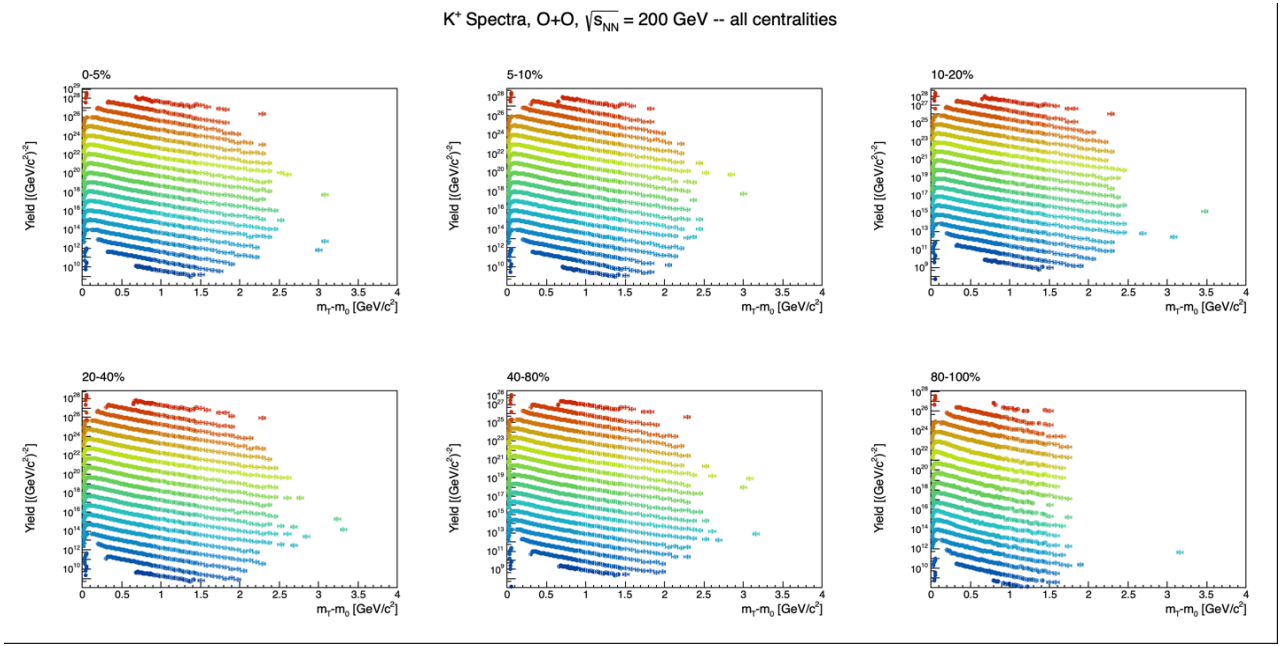
BTOF-corrected pion+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Pion-: Combined All-Centralities Spectra View



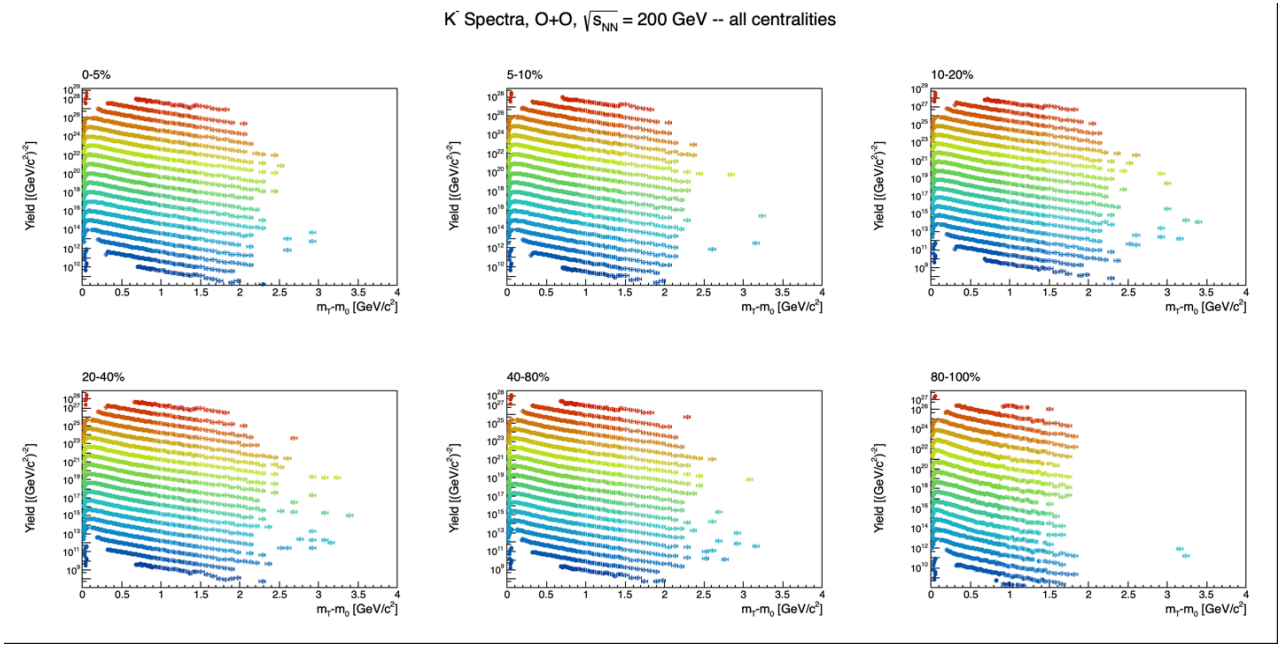
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Kaon+: Combined All-Centralities Spectra View



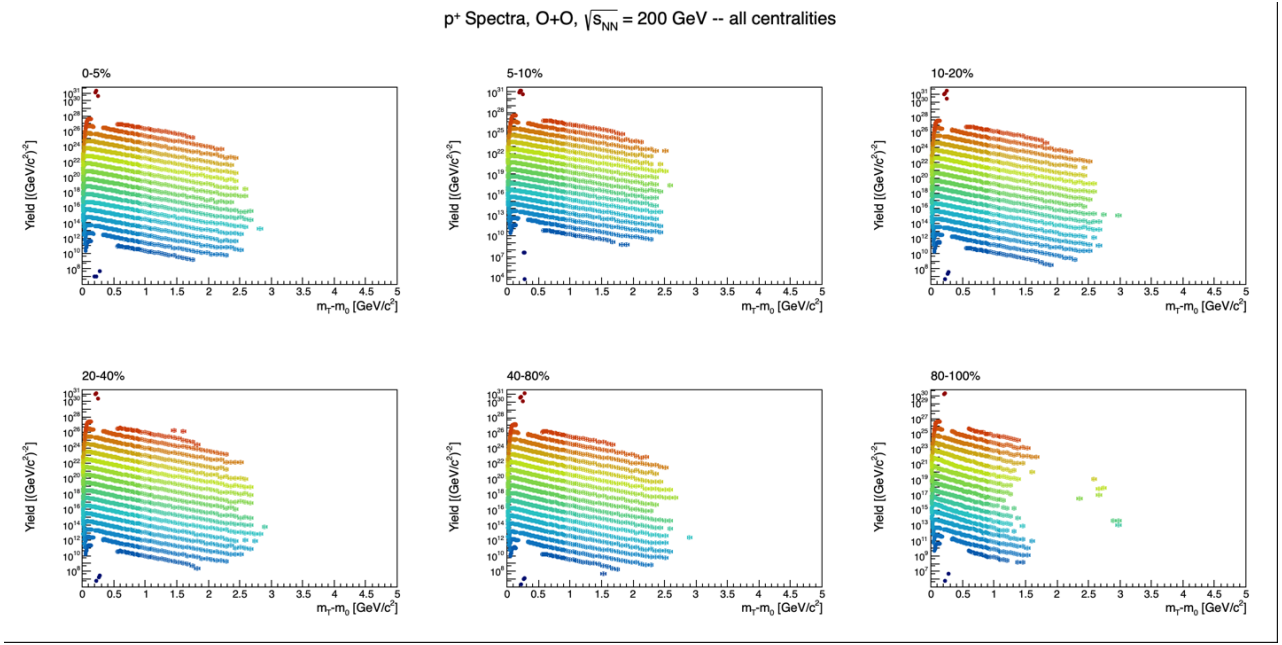
BTOF-corrected kaon+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Kaon-: Combined All-Centralities Spectra View



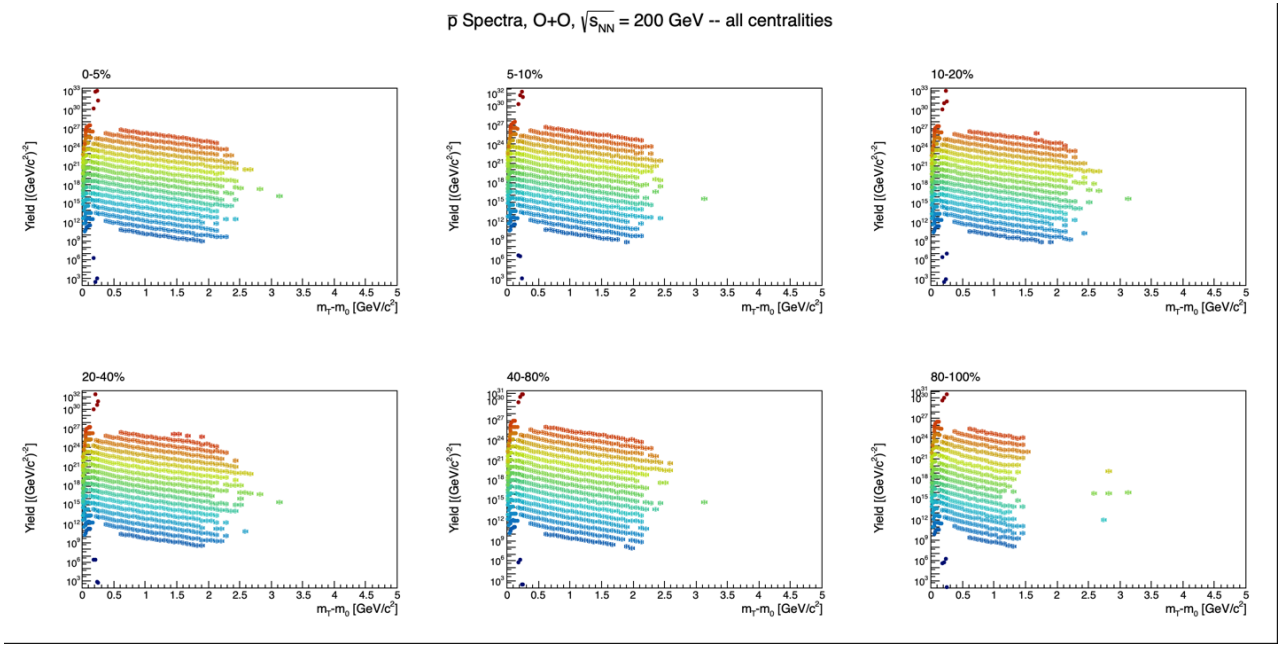
BTOF-corrected kaon- mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Proton+: Combined All-Centralities Spectra View



BTOF-corrected proton+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Antiproton: All-Centralities Spectra View



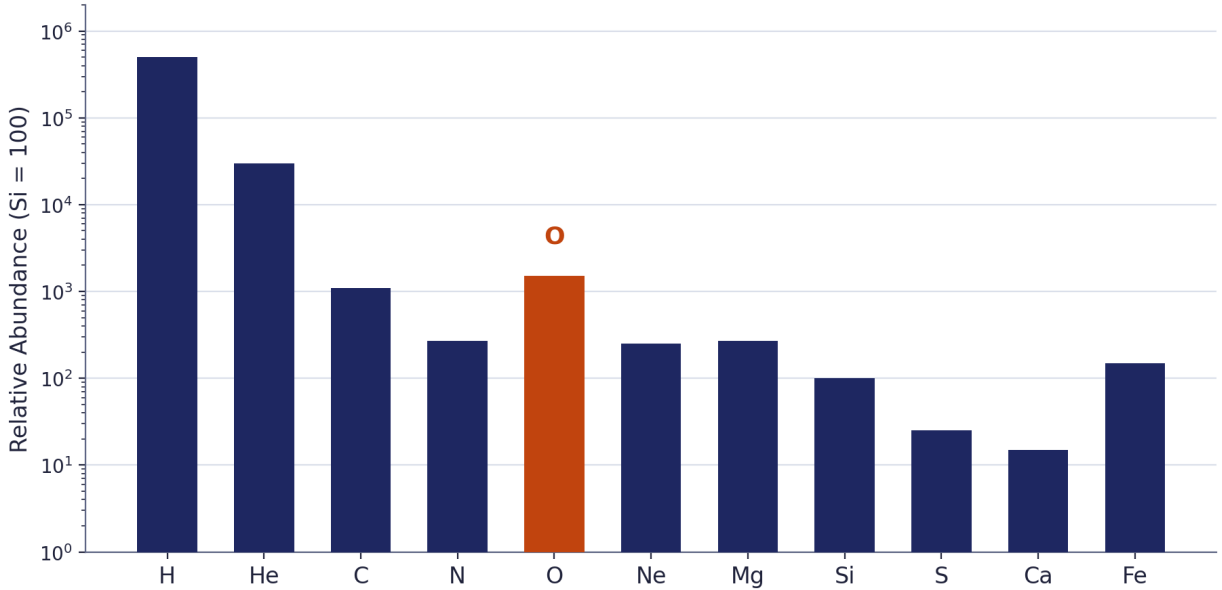
BTOF-corrected antiproton mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Summary & Next Steps

- **Motivation** -- O+O collisions as a galactic cosmic ray proxy for space radiation and shielding studies.
- **BTOF-corrected spectra produced for π/K/p** -- all six centrality bins, TPC+BTOF combined.
- **Currently working on eTOF calibrations** -- MuDsts restored, calibration pass underway; this would be the first use of eTOF at 200 GeV.
- **Applying the P23ic embedding** -- TPC efficiency, energy loss, and hybrid feed-down will all use the existing P23ic/SL23c embedding.
- **Open question** -- does an StRefMultCorr exist for O+O in the SL23c library? Previous O+O analyses have used the EPD instead of refmult for centrality determination.

BACKUP

GCR Elemental Composition

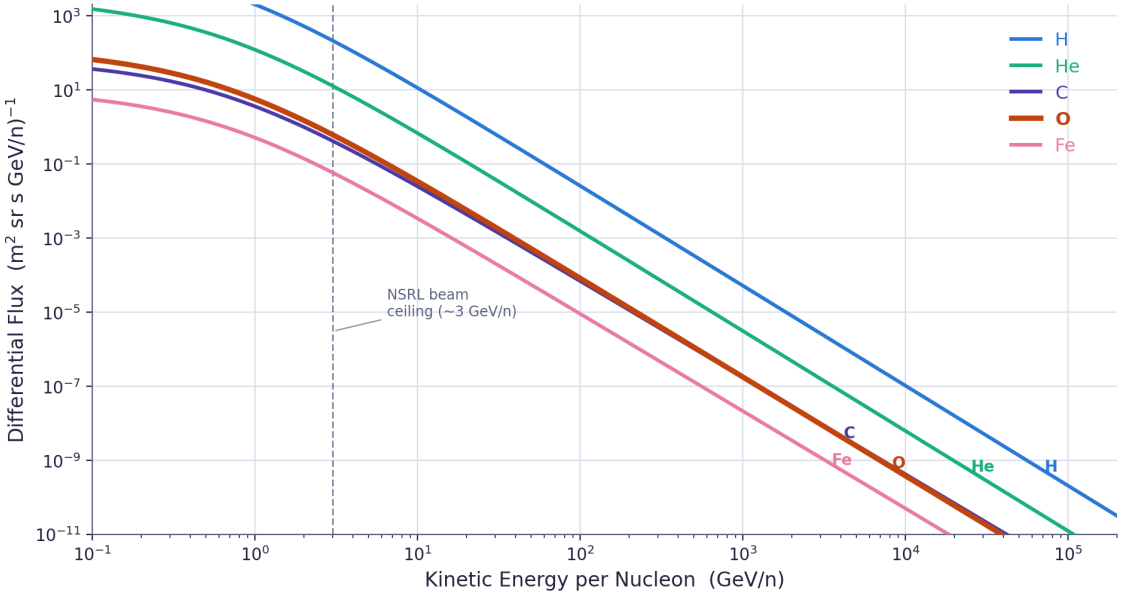


Oxygen is a major primary GCR component and spallation parent -- directly relevant to the O+O system studied here.

Source: Simpson (1983), Ann. Rev. Nucl. Part. Sci. 33:323; ACE-CRIS (Caltech ACE News #83). Representative reconstruction, not a digitized figure.

BACKUP

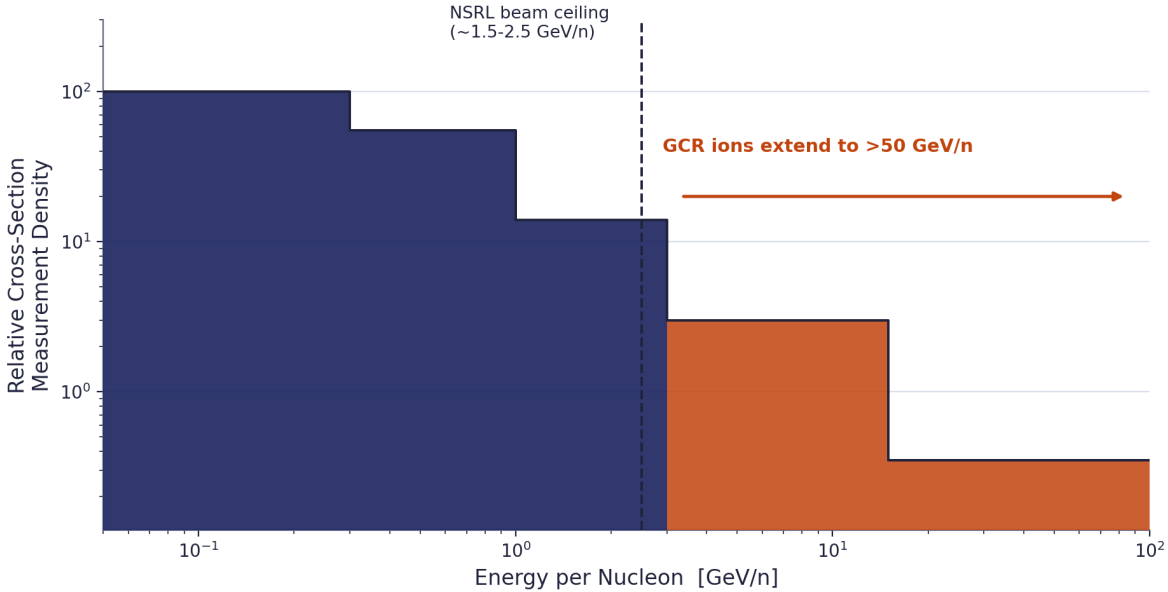
GCR Energy Spectrum by Element



Sources: PDG "Cosmic Rays" review; Ahn et al. 2009, ApJ 707:593 (CREAM); Simpson (1983) abundances. Representative reconstruction, not a digitized figure.

BACKUP

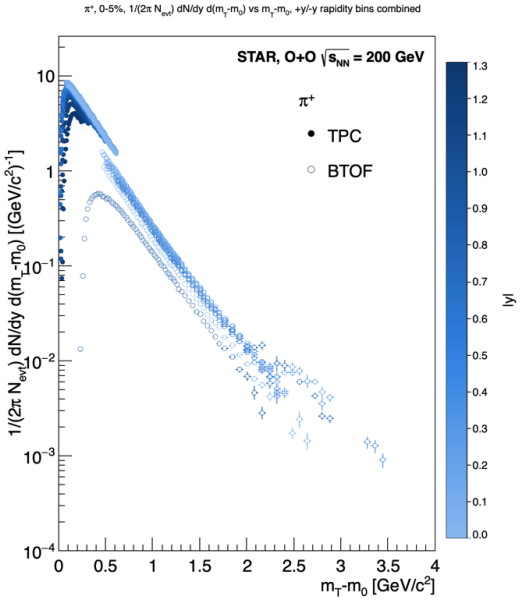
Nuclear Data Coverage Gap



FLUKA/GEANT4/PHITS are tuned to measured cross sections -- data thins sharply above ~3 GeV/n, beyond NSRL's beam ceiling.

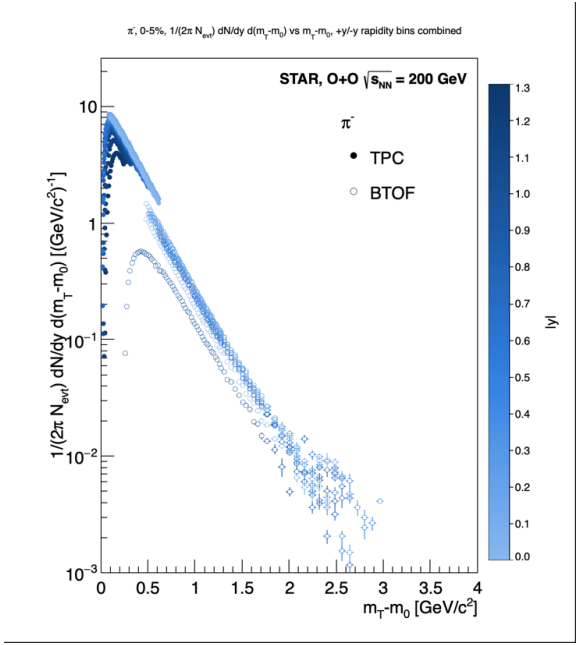
Source: Norbury & Losekamm, NASA NTRS 20240015733; NSRL beam specs, BNL. Representative reconstruction, not a digitized figure.

Raw Yield Extraction: Z-Fitter Output (π+)



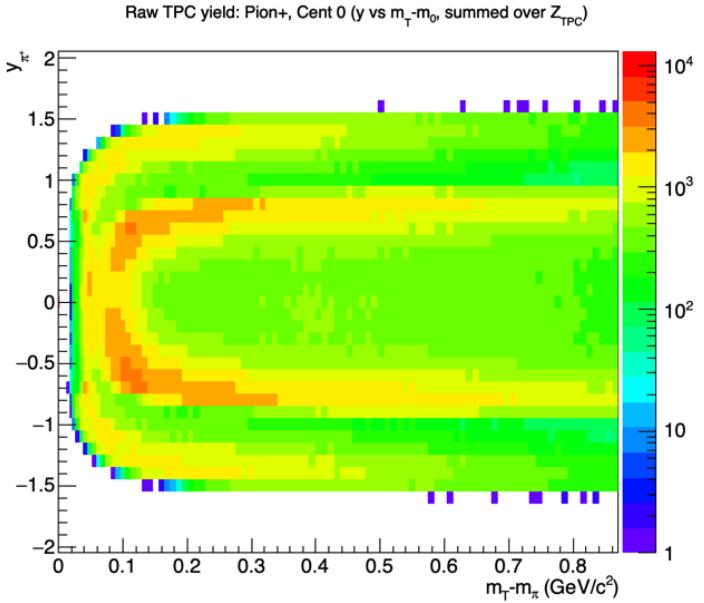
TPC vs BTOF raw dN/dy vs mT-m0 for pion+, 0-5% centrality -- before the two detectors are merged and corrected.

Raw Yield Extraction: Z-Fitter Output (π−)



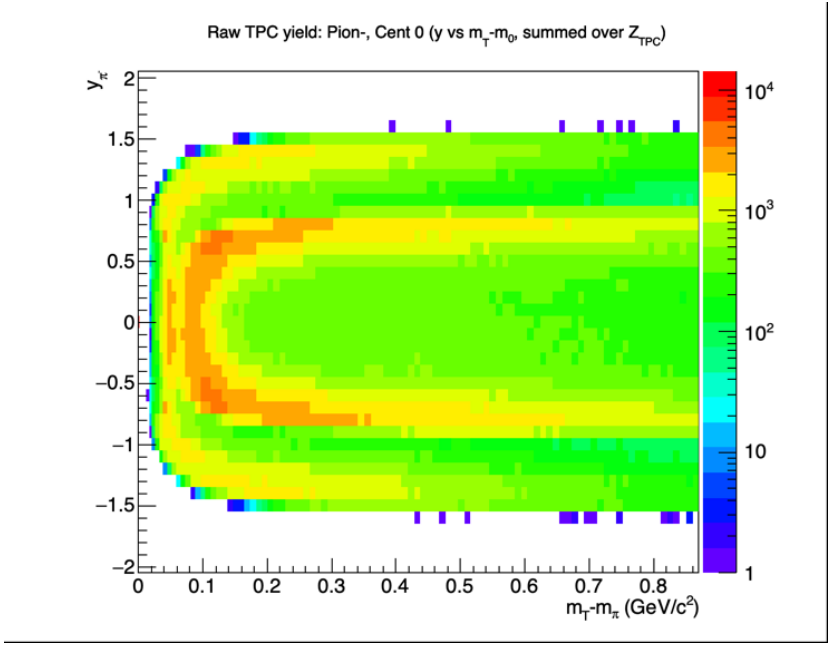
TPC vs BTOF raw dN/dy vs mT-m0 for pion-, 0-5% centrality -- before the two detectors are merged and corrected.

Raw TPC Yield: Rapidity vs mT-m0 (π+)



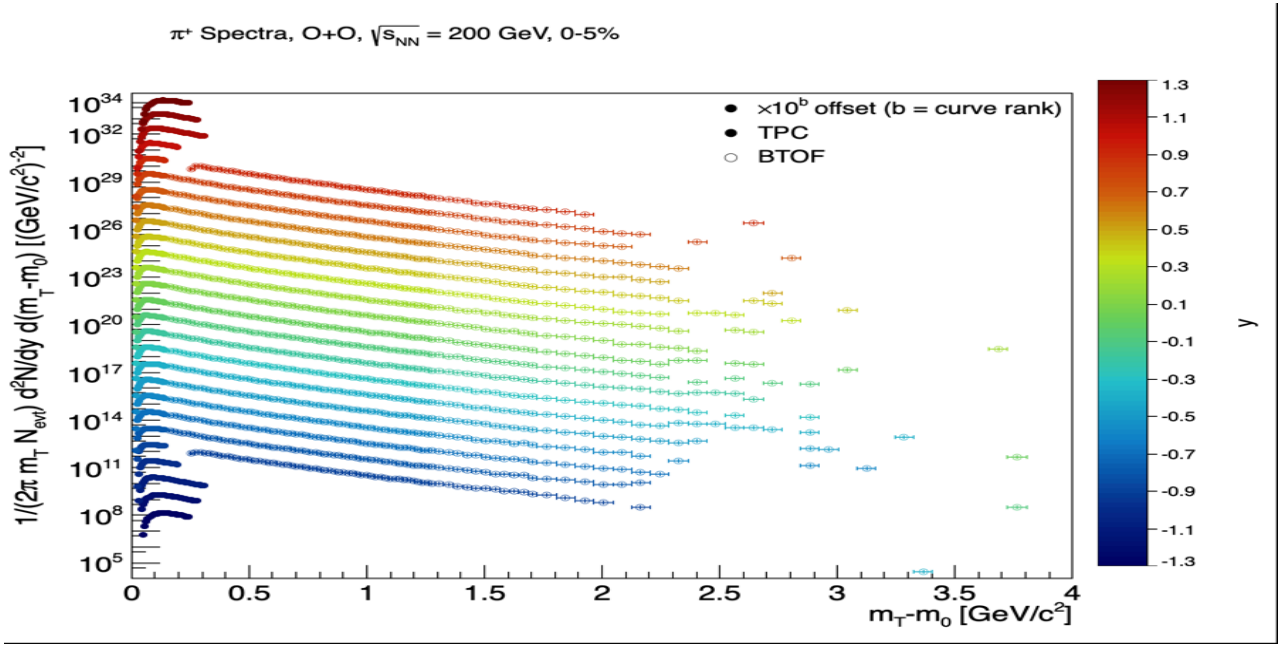
Pion+ raw TPC yield vs rapidity and mT-m0, 0-5% centrality, summed over Z_TPC -- input to the corrections.

Raw TPC Yield: Rapidity vs mT-m0 (π−)



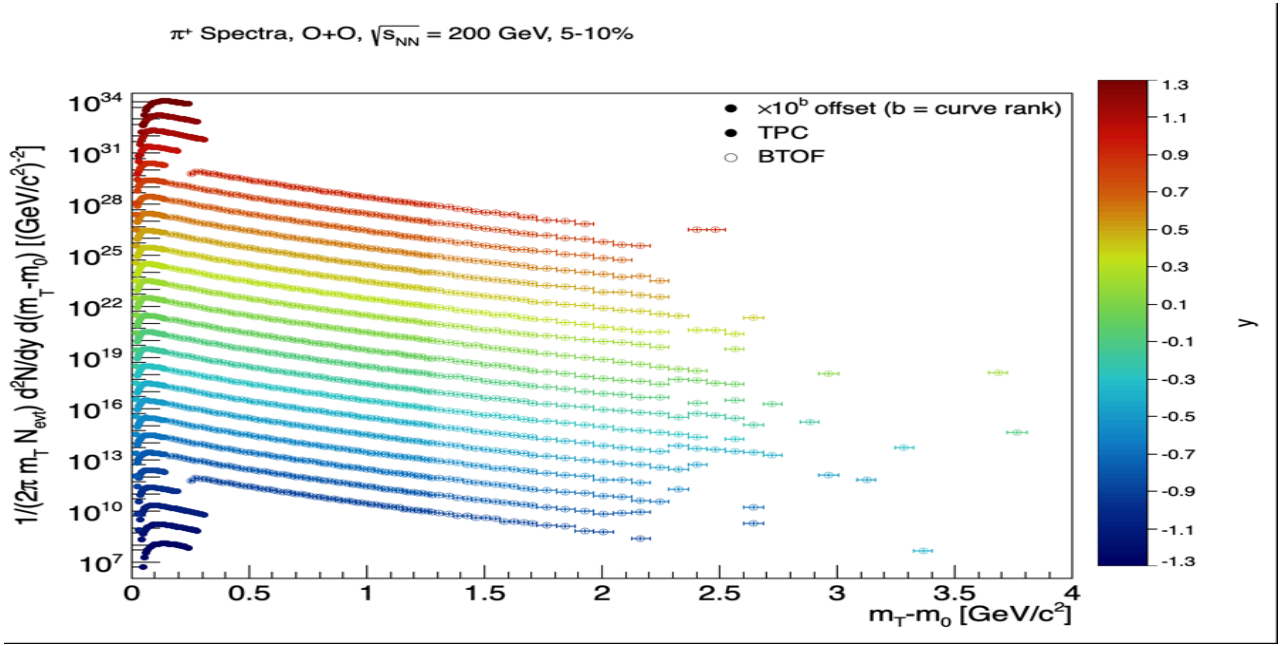
Pion- raw TPC yield vs rapidity and mT-m0, 0-5% centrality, summed over Z_TPC -- input to the corrections.

Pion+: 0-5% Centrality



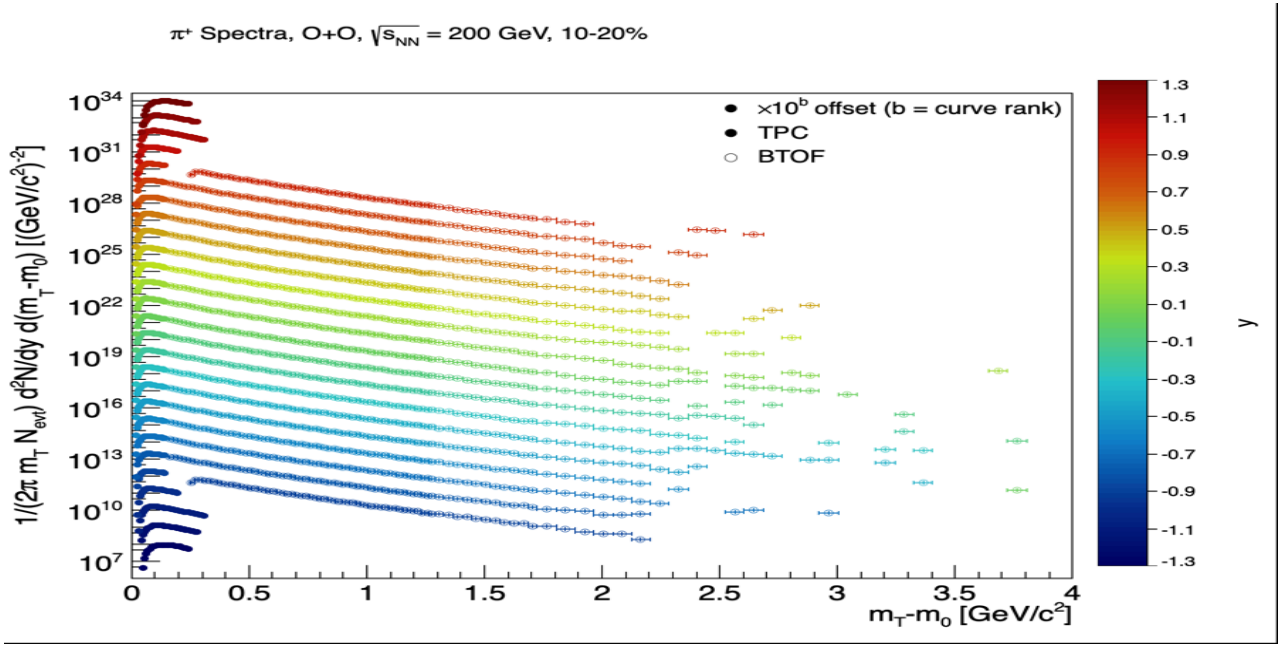
BTOF-corrected pion+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Pion+: 5-10% Centrality



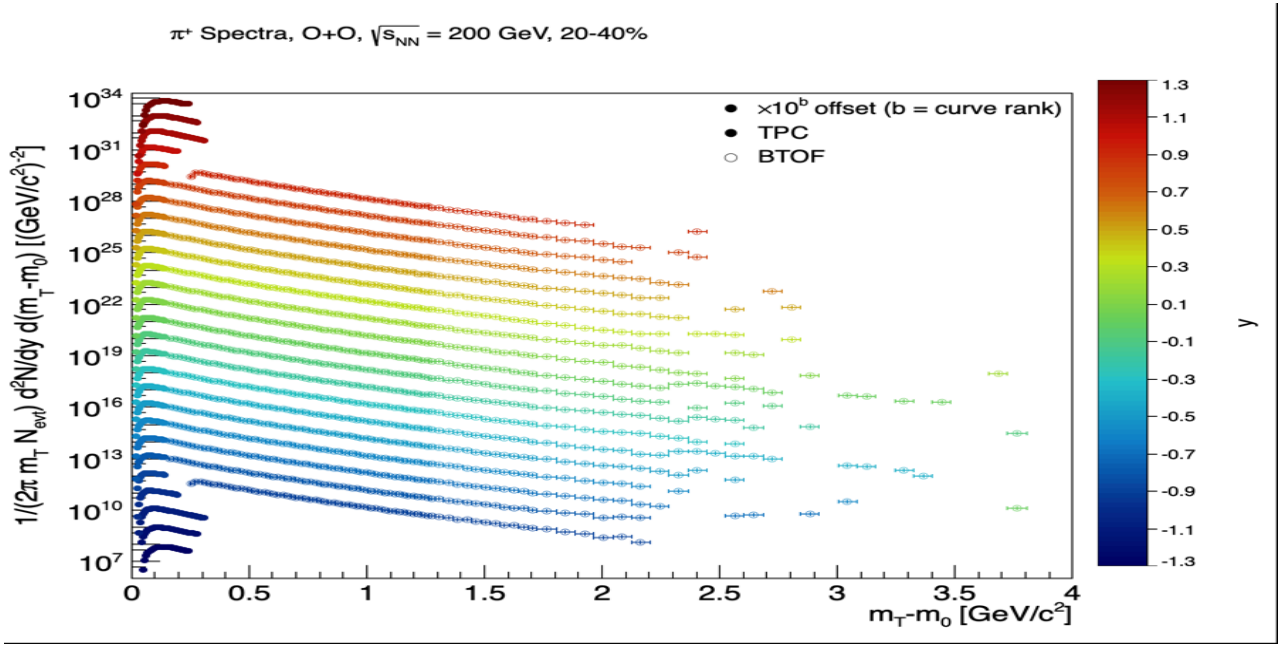
BTOF-corrected pion+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Pion+: 10-20% Centrality



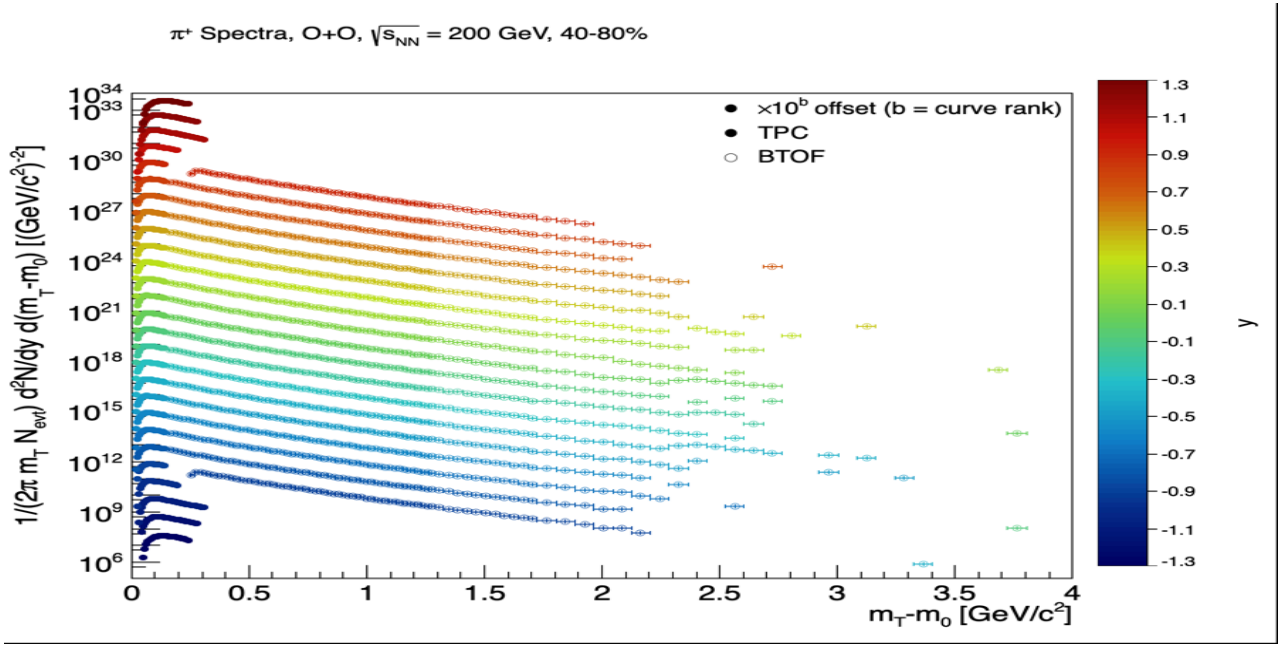
BTOF-corrected pion+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Pion+: 20-40% Centrality



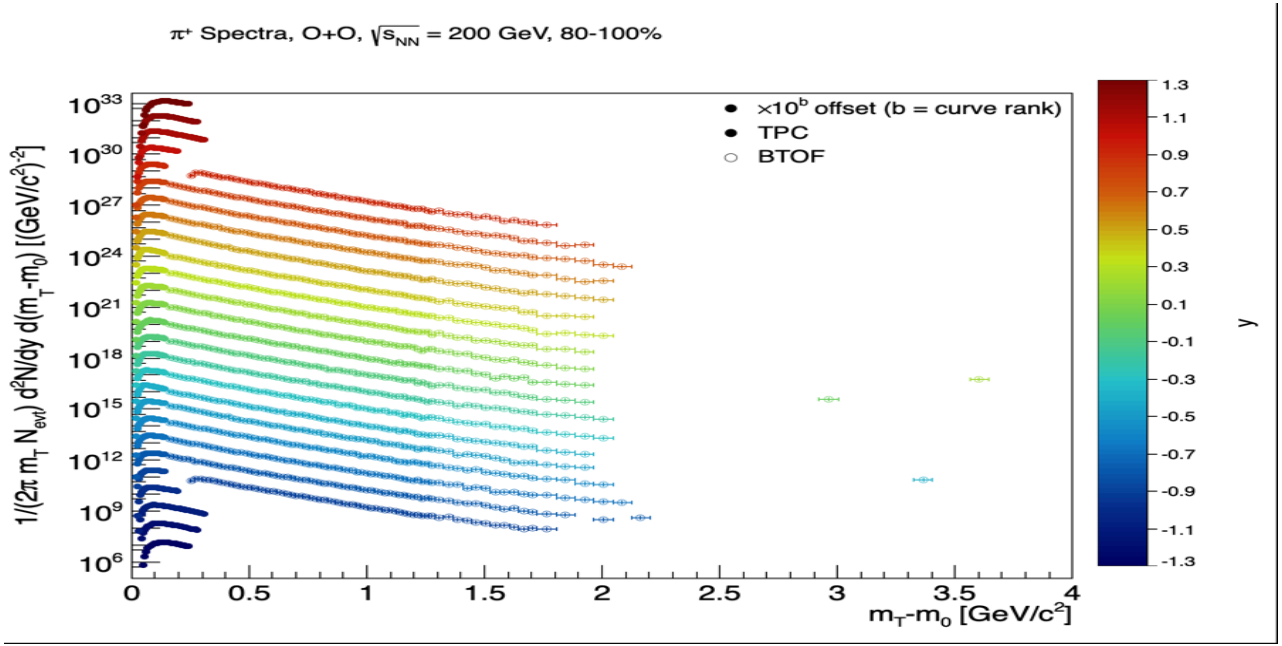
BTOF-corrected pion+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Pion+: 40-80% Centrality



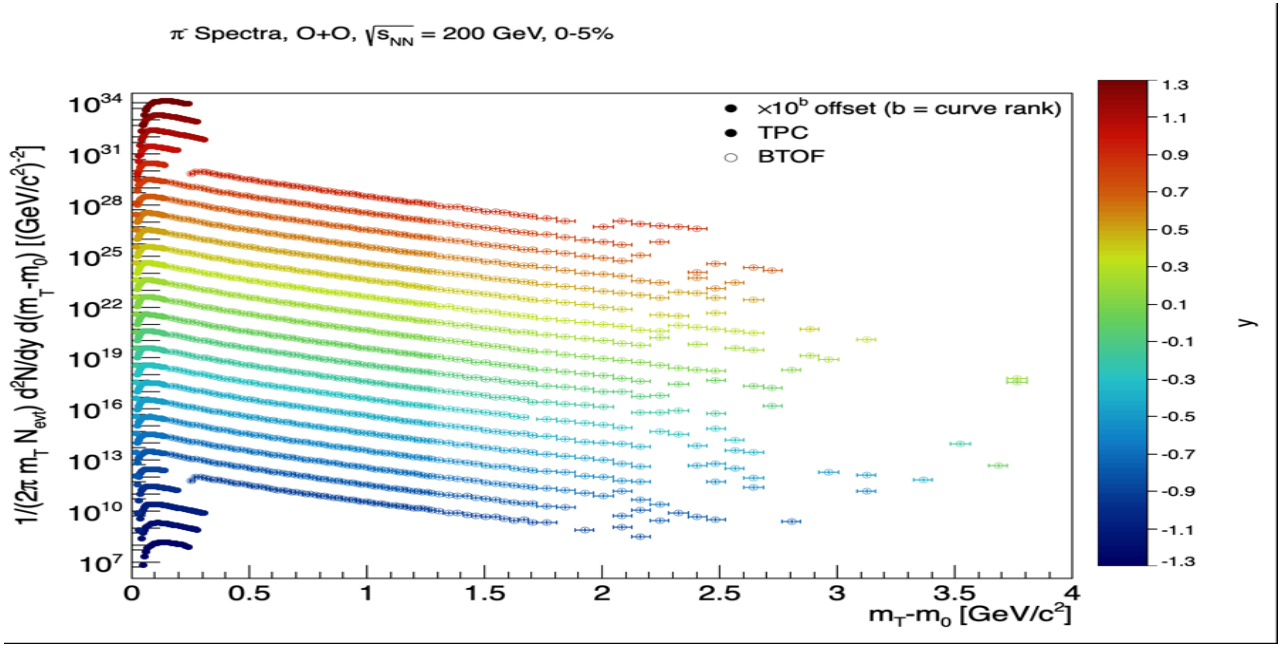
BTOF-corrected pion+ mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Pion+: 80-100% Centrality



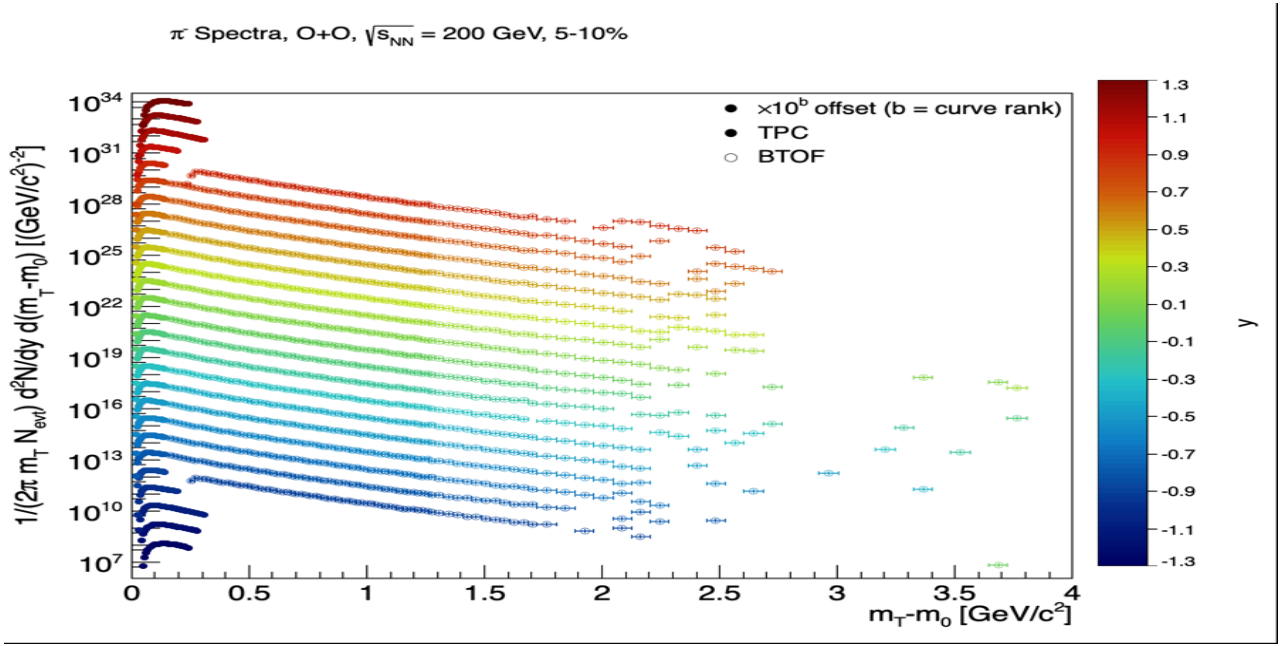
BTOF-corrected pion+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Pion-: 0-5% Centrality



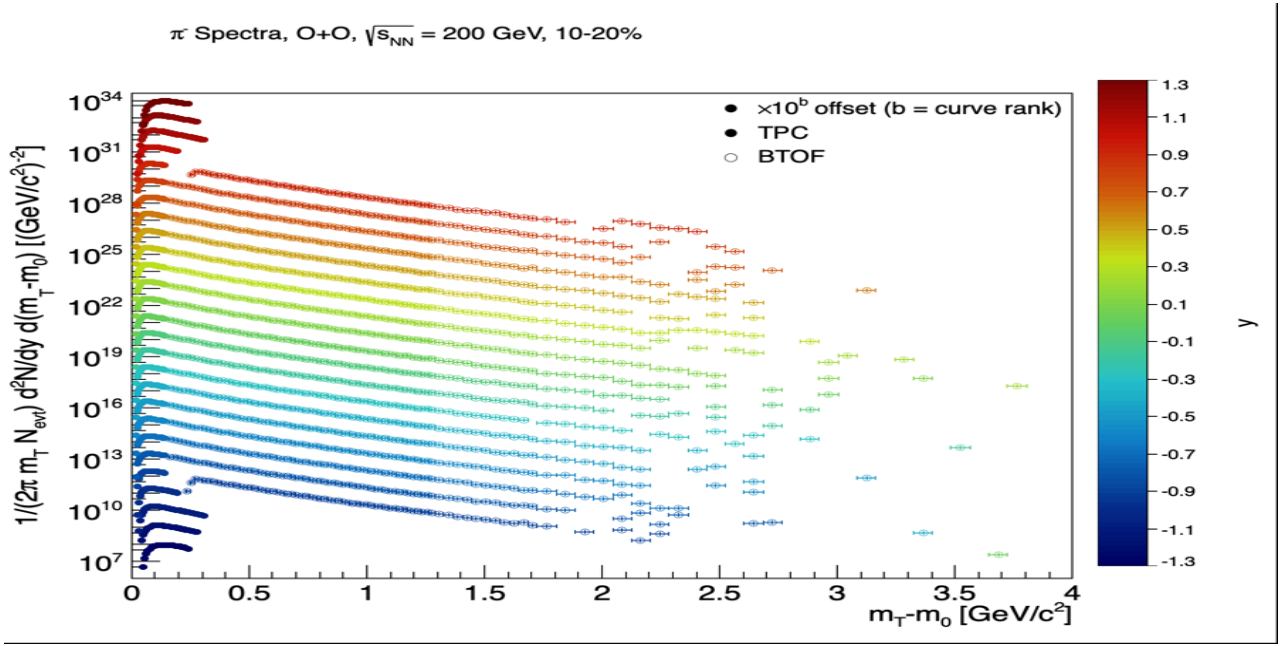
BTOF-corrected pion- mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Pion-: 5-10% Centrality



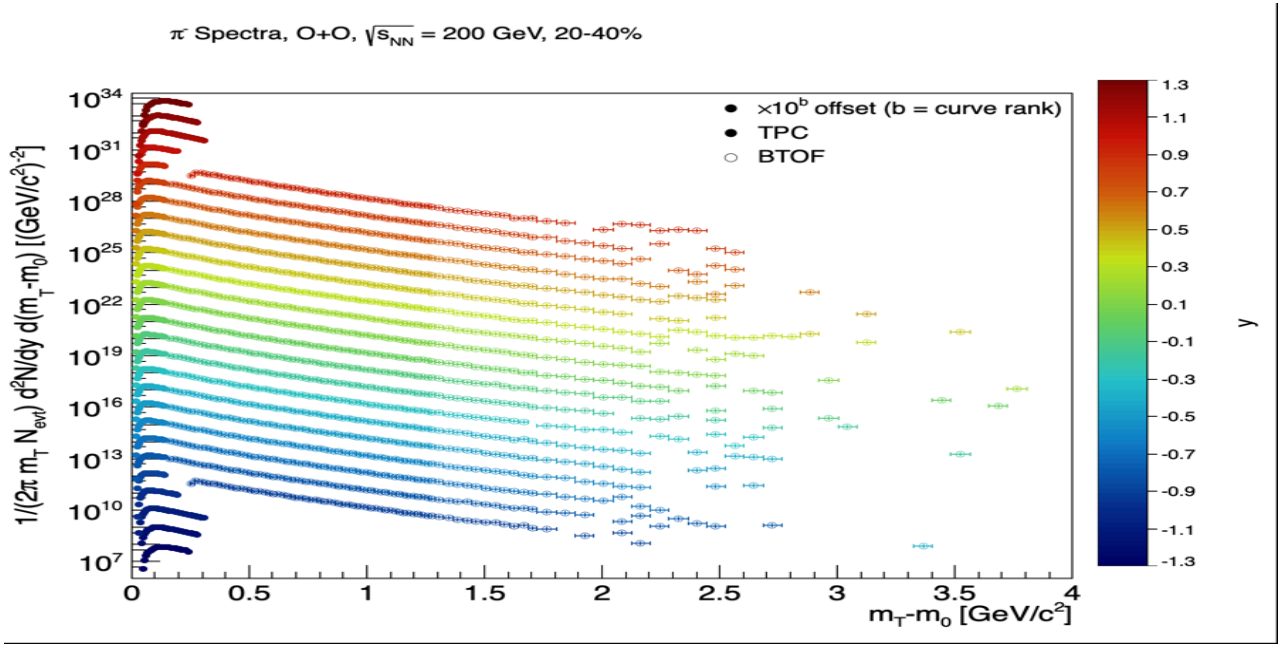
BTOF-corrected pion- mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Pion-: 10-20% Centrality



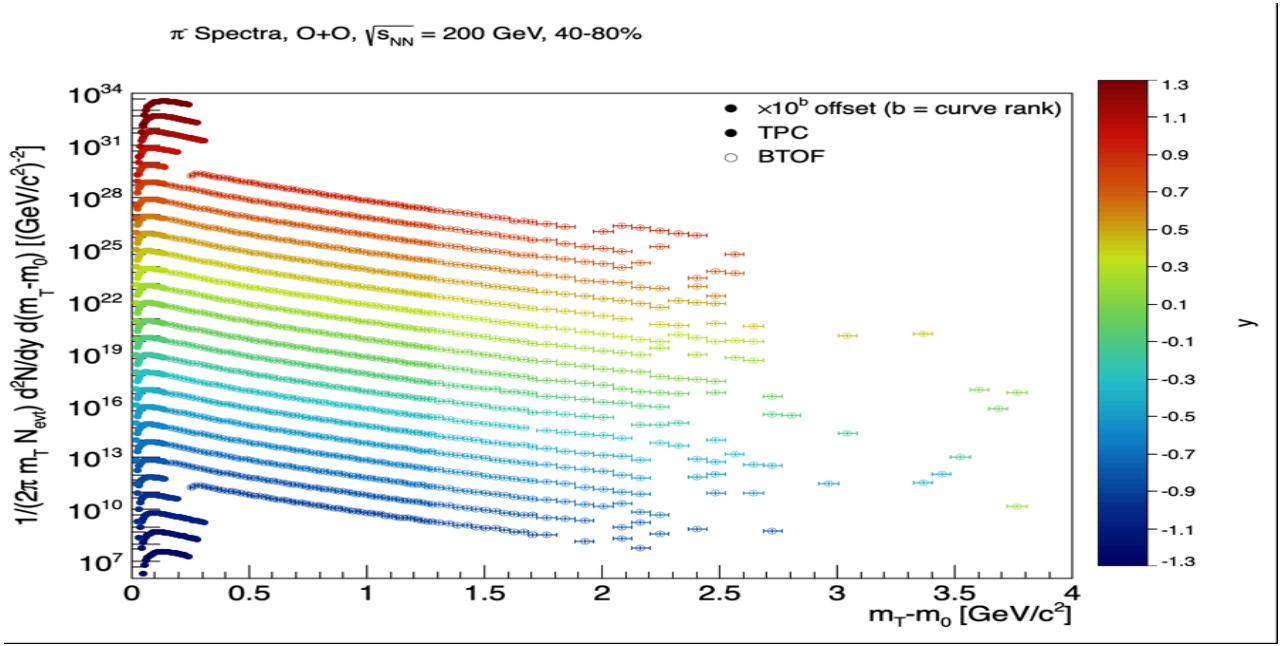
BTOF-corrected pion- mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Pion-: 20-40% Centrality



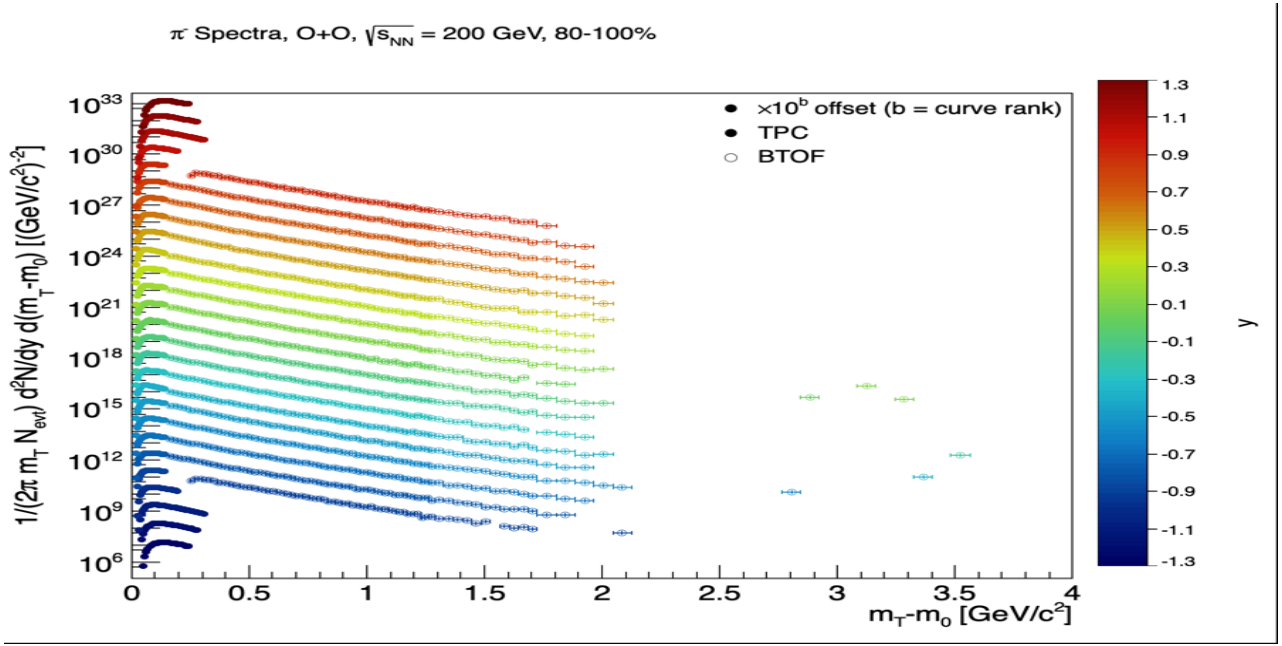
BTOF-corrected pion- mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Pion-: 40-80% Centrality



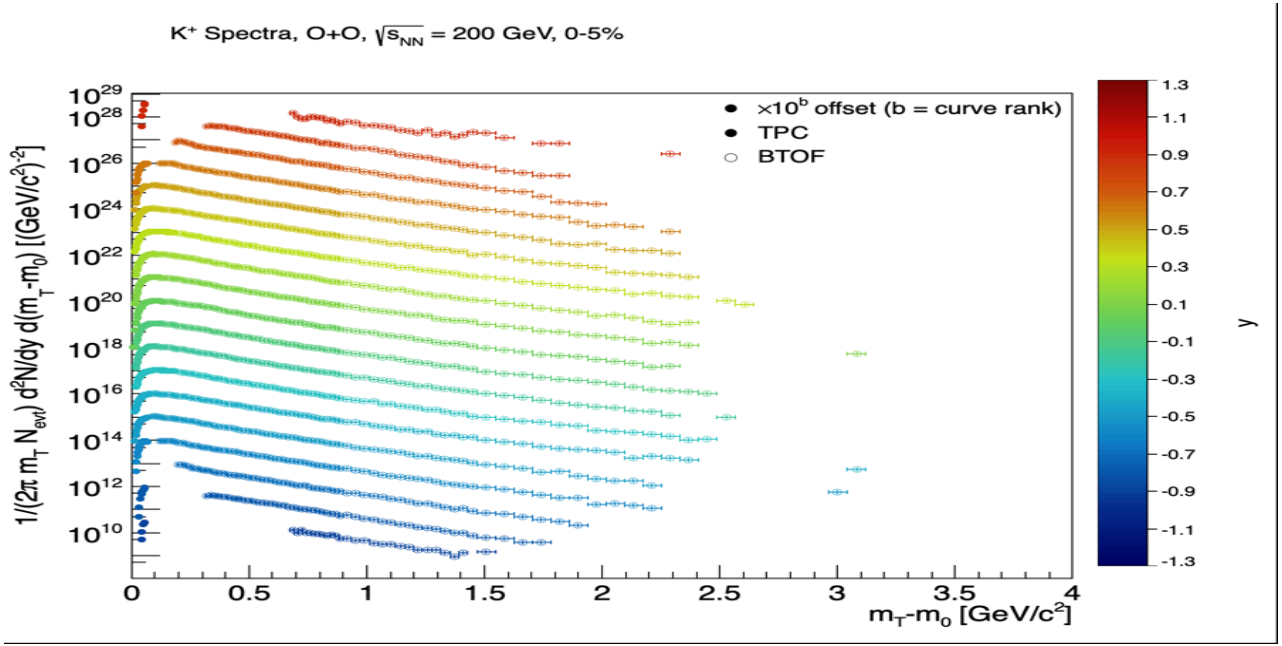
BTOF-corrected pion- mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Pion-: 80-100% Centrality



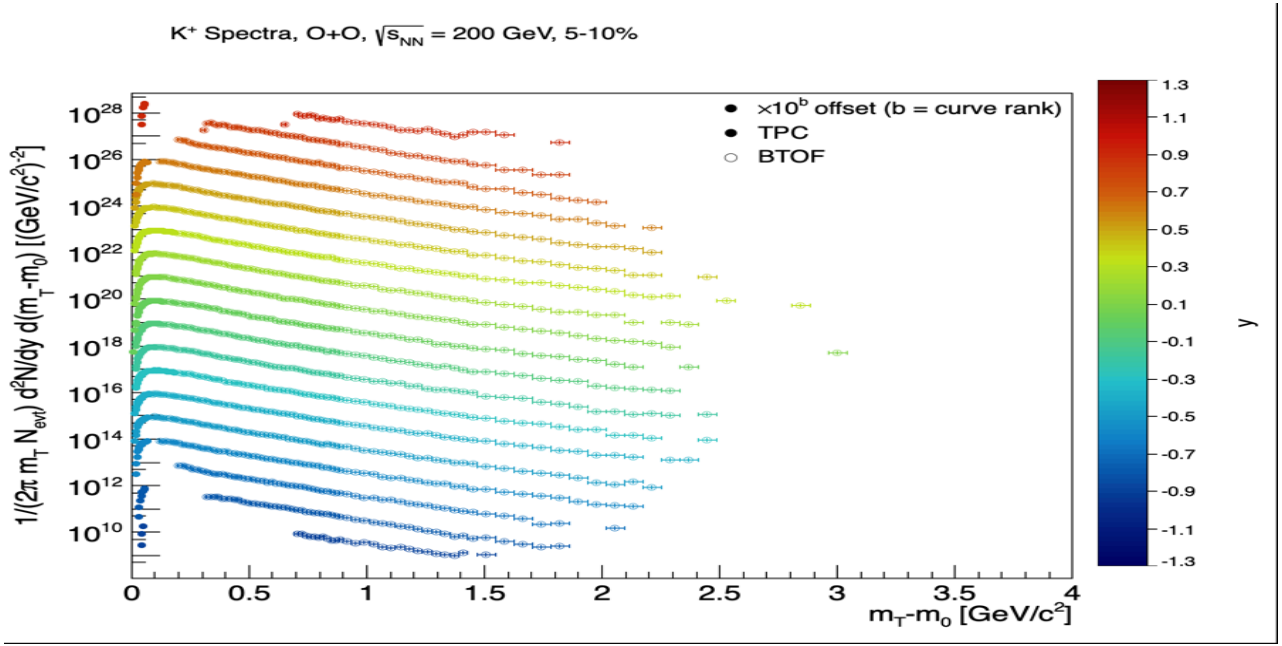
BTOF-corrected pion- mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 0-5% Centrality



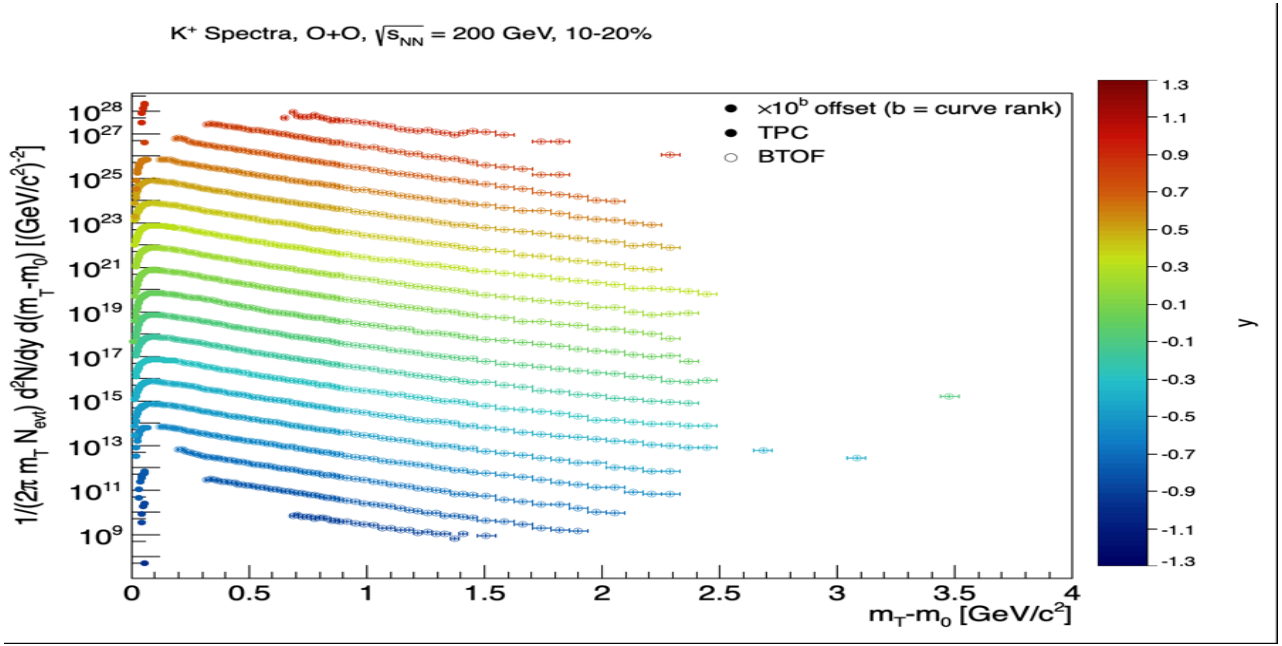
BTOF-corrected kaon+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 5-10% Centrality



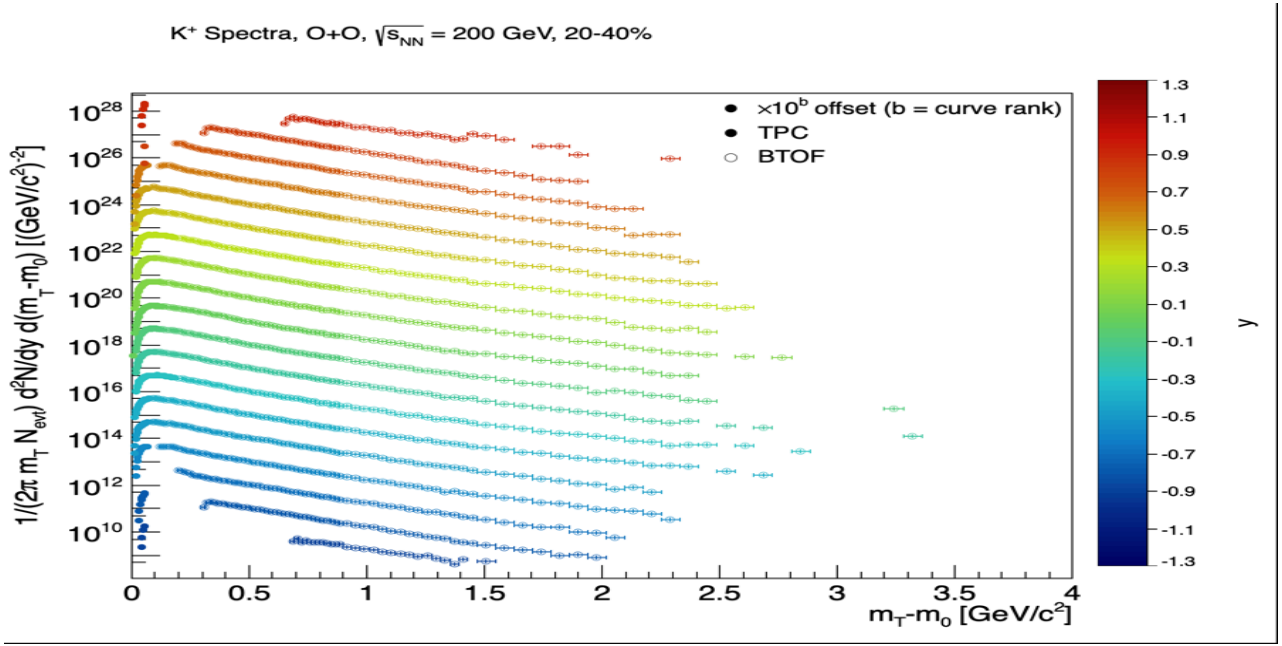
BTOF-corrected kaon+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 10-20% Centrality



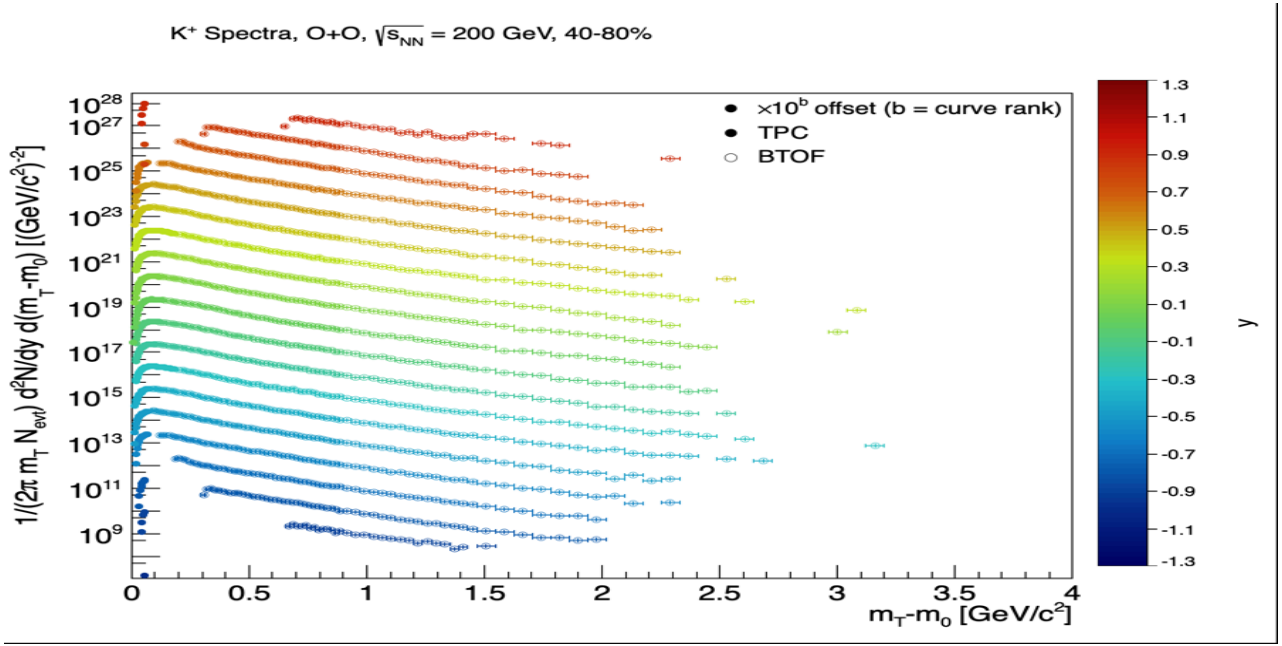
BTOF-corrected kaon+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 20-40% Centrality



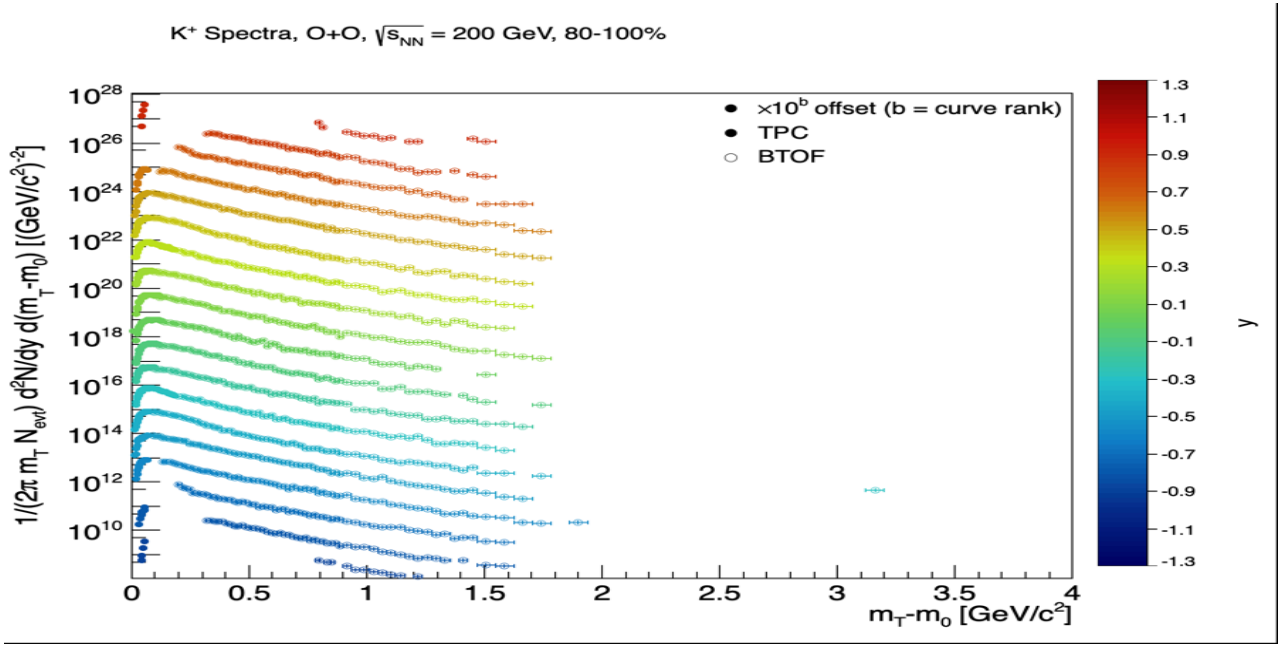
BTOF-corrected kaon+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 40-80% Centrality



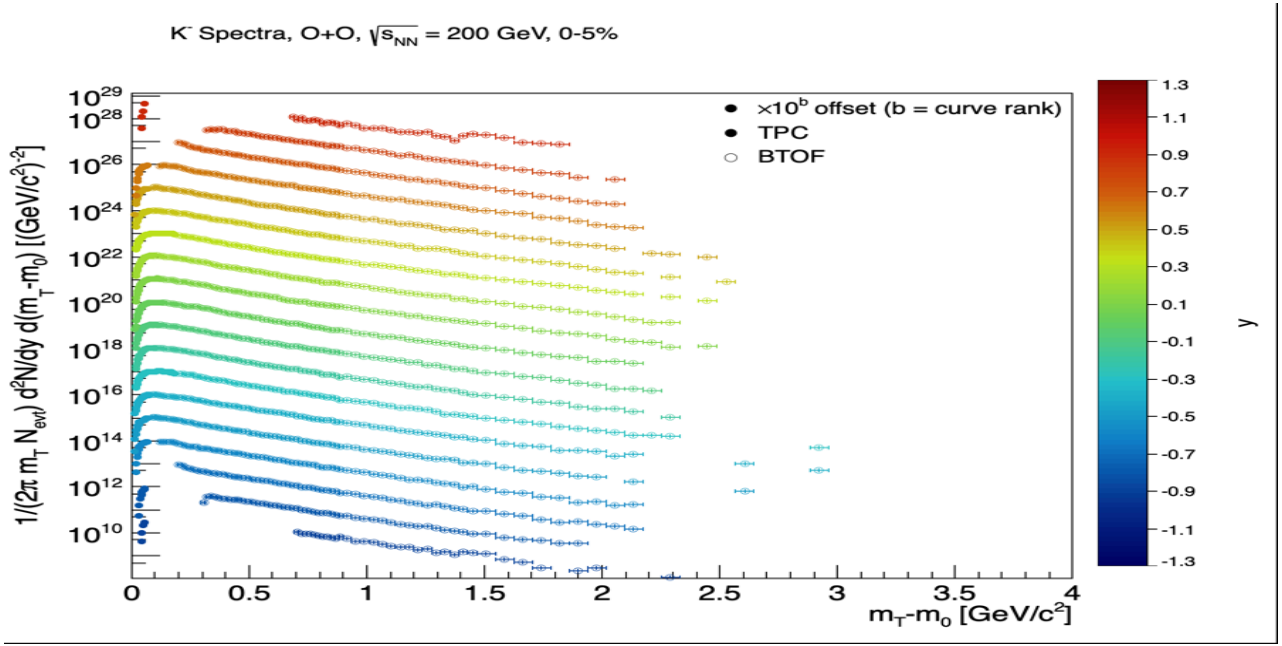
BTOF-corrected kaon+ mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 80-100% Centrality



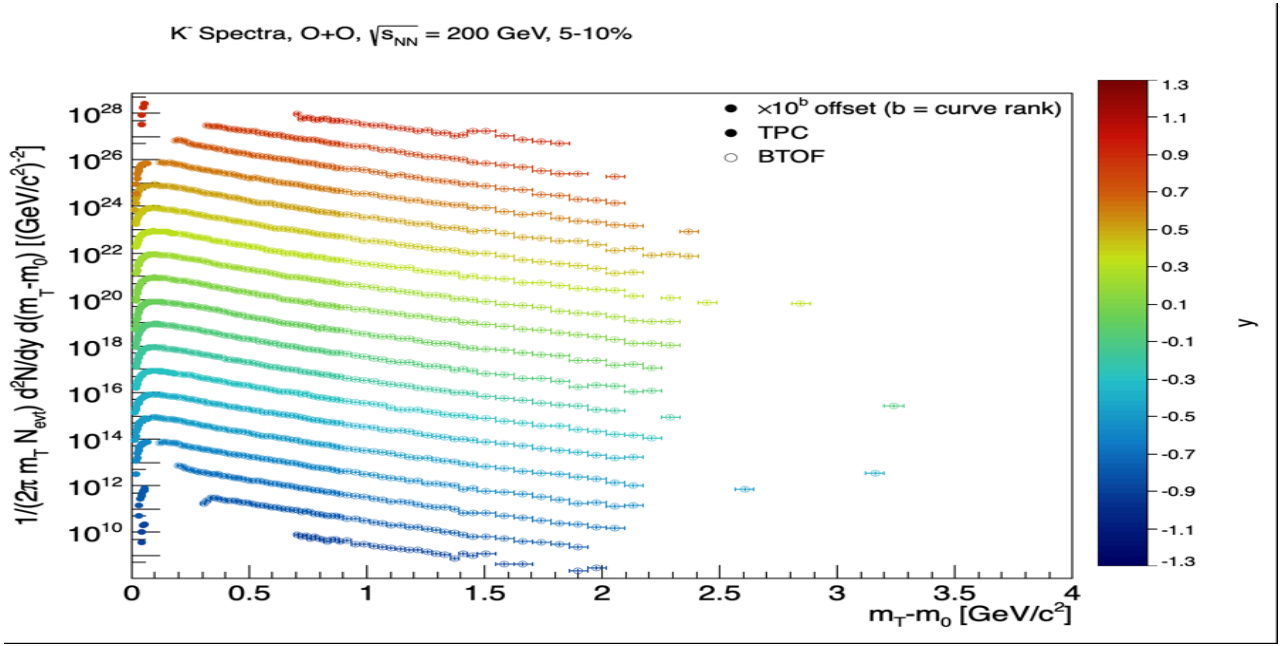
BTOF-corrected kaon+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 0-5% Centrality



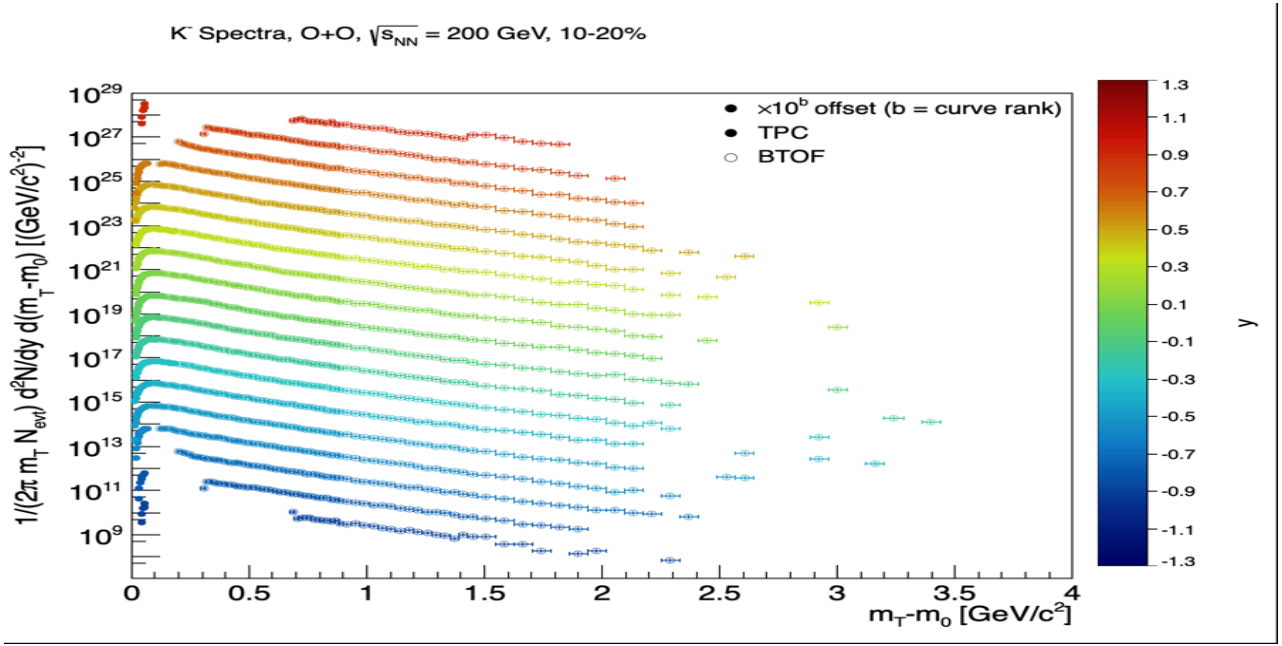
BTOF-corrected kaon- mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 5-10% Centrality



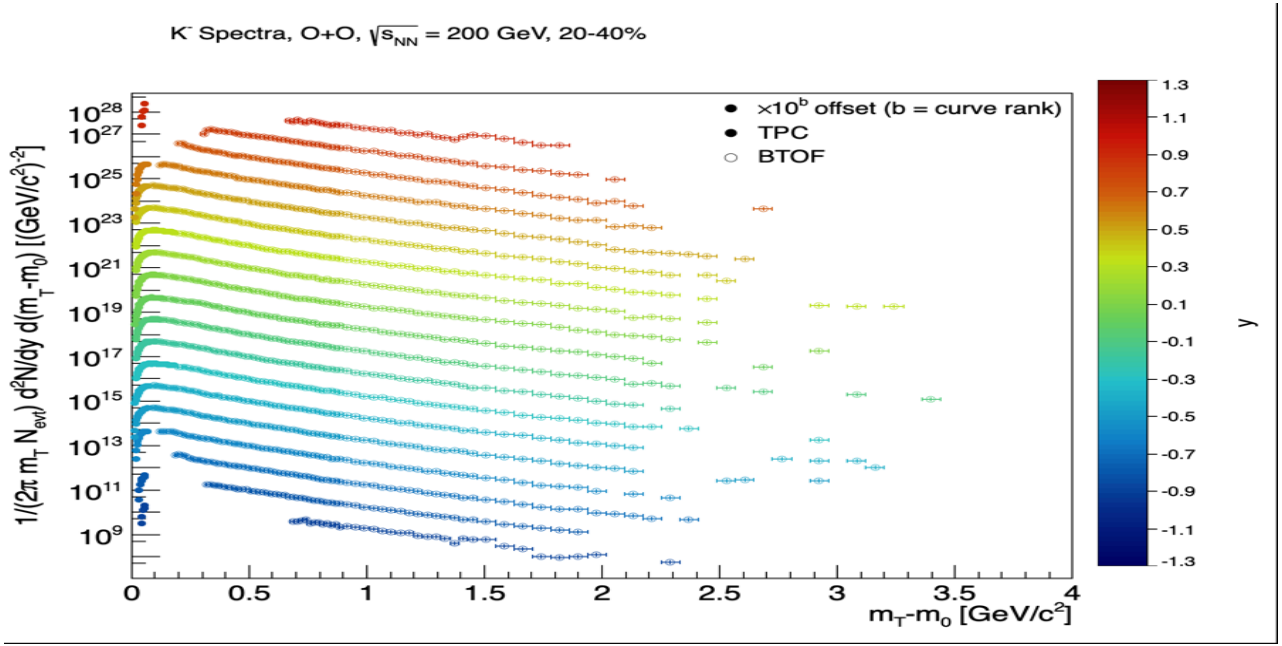
BTOF-corrected kaon- mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 10-20% Centrality



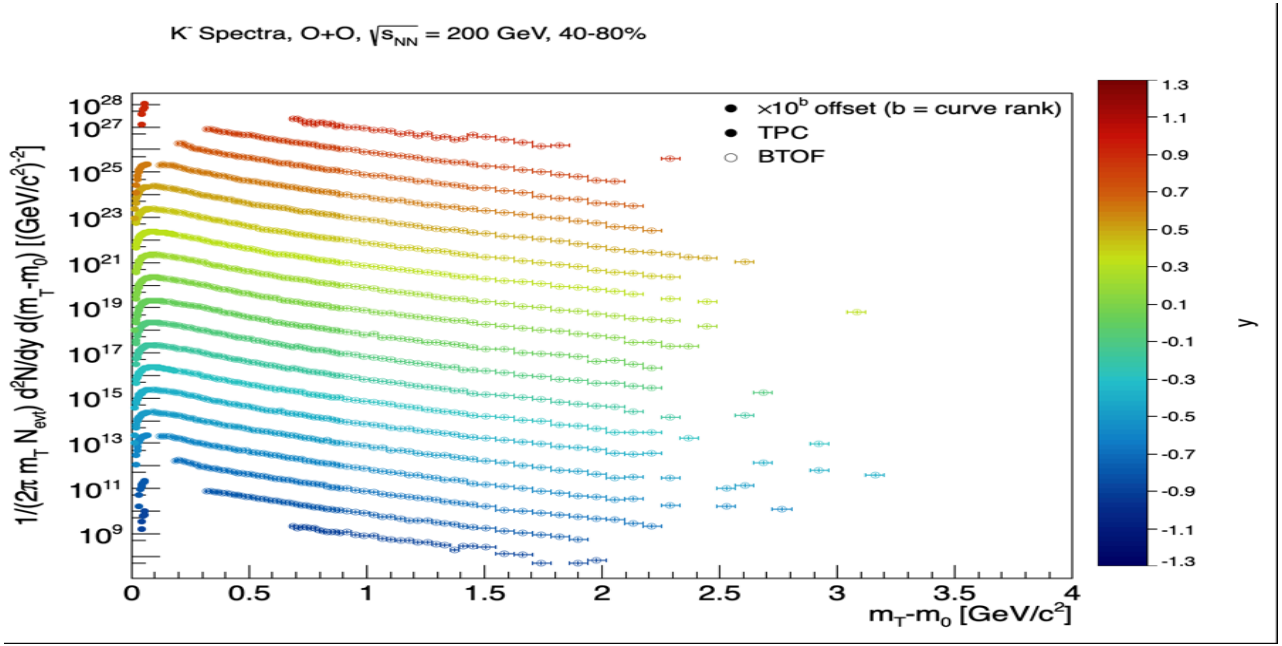
BTOF-corrected kaon- mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 20-40% Centrality



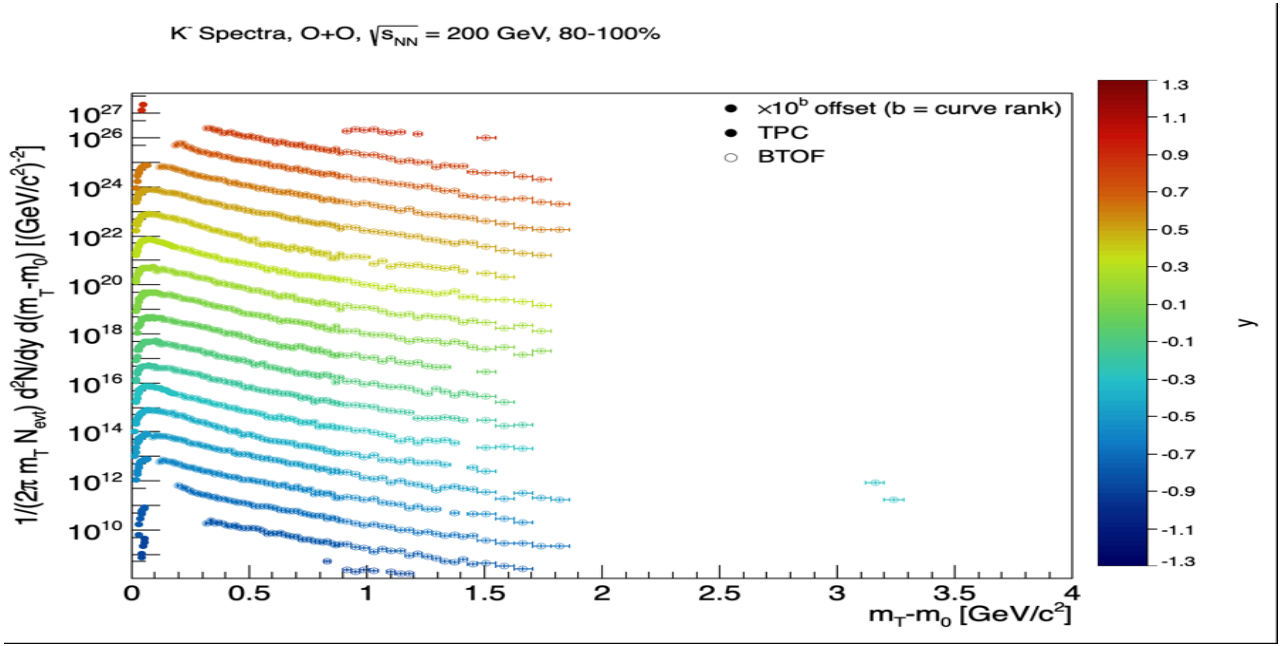
BTOF-corrected kaon- mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 40-80% Centrality



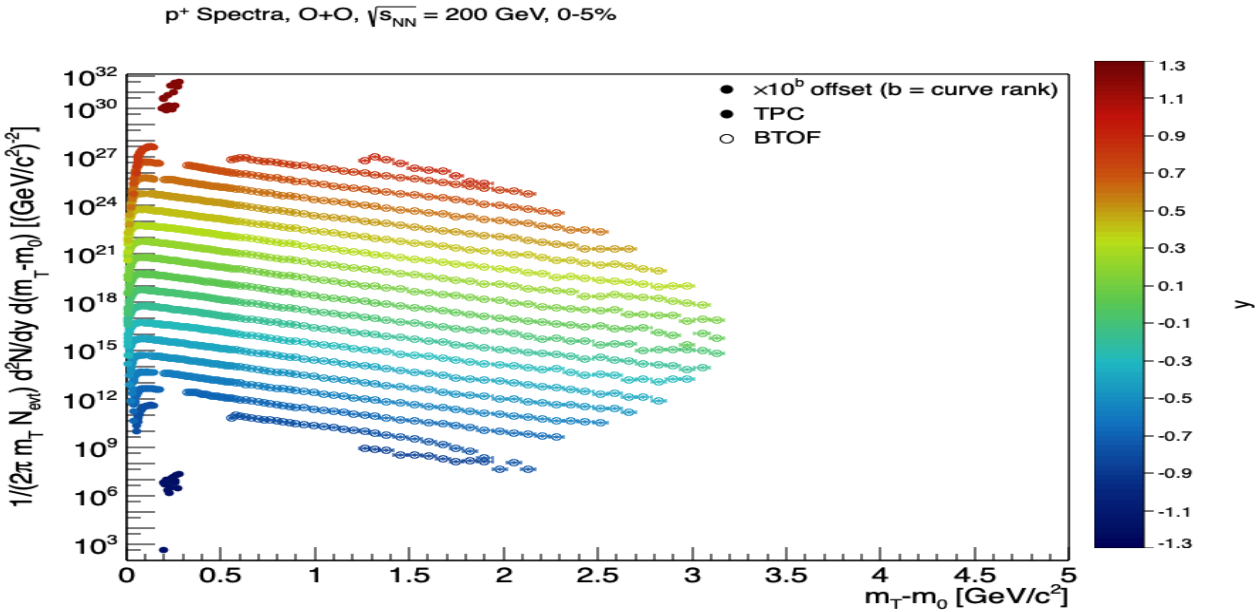
BTOF-corrected kaon- mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 80-100% Centrality



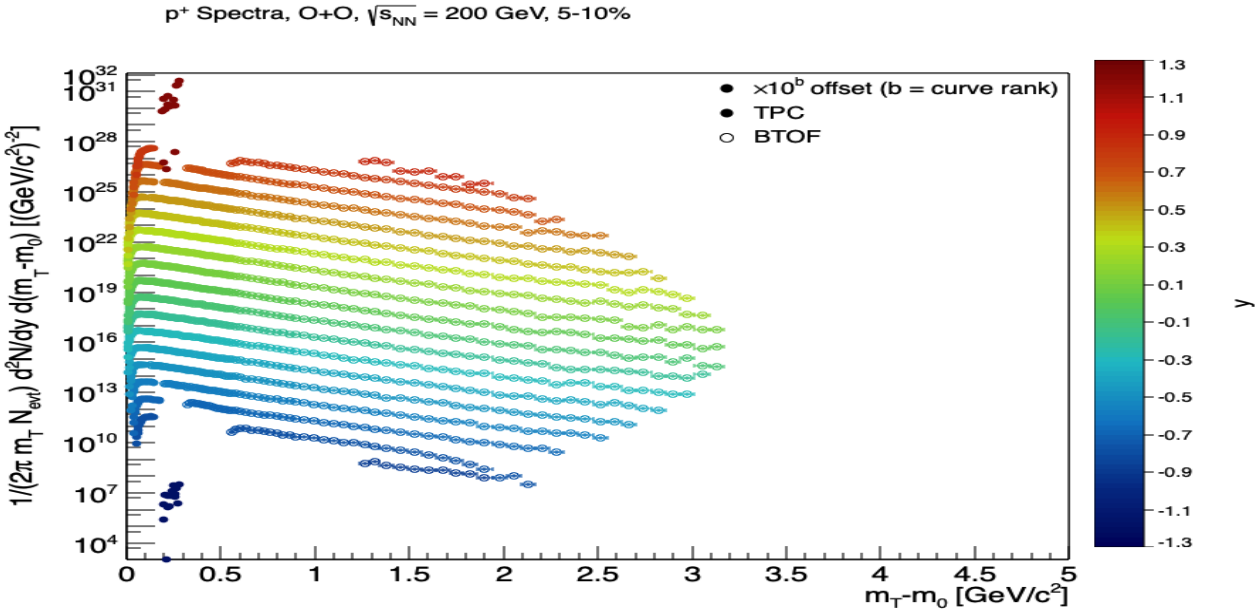
BTOF-corrected kaon- mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Proton+: 0-5% Centrality



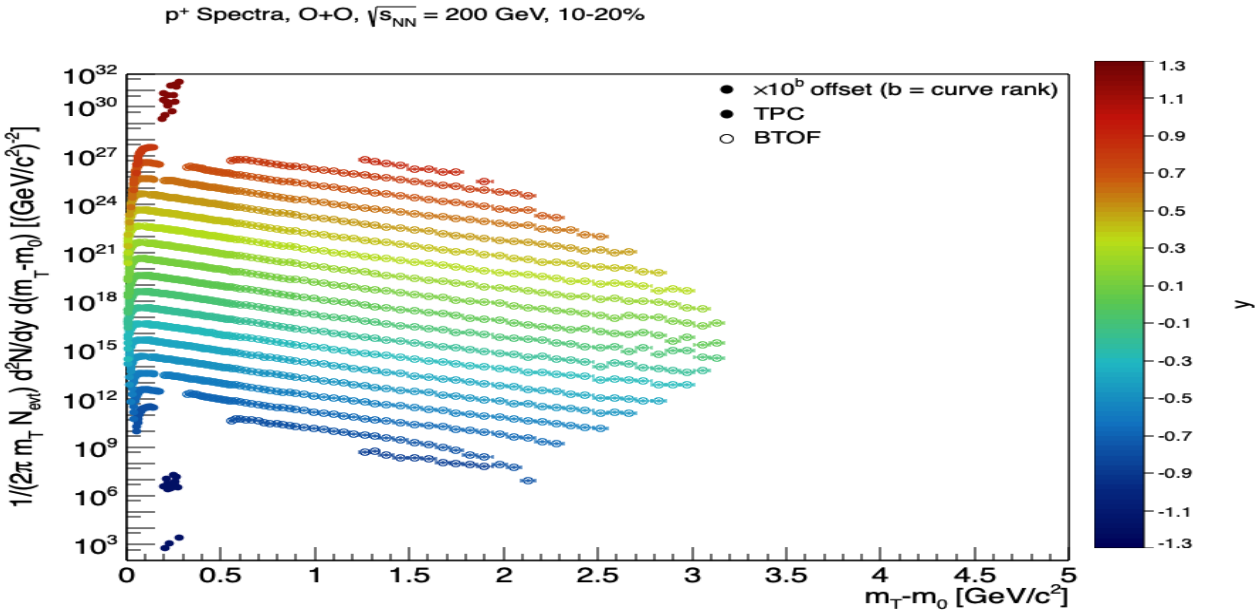
BTOF-corrected proton+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Proton+: 5-10% Centrality



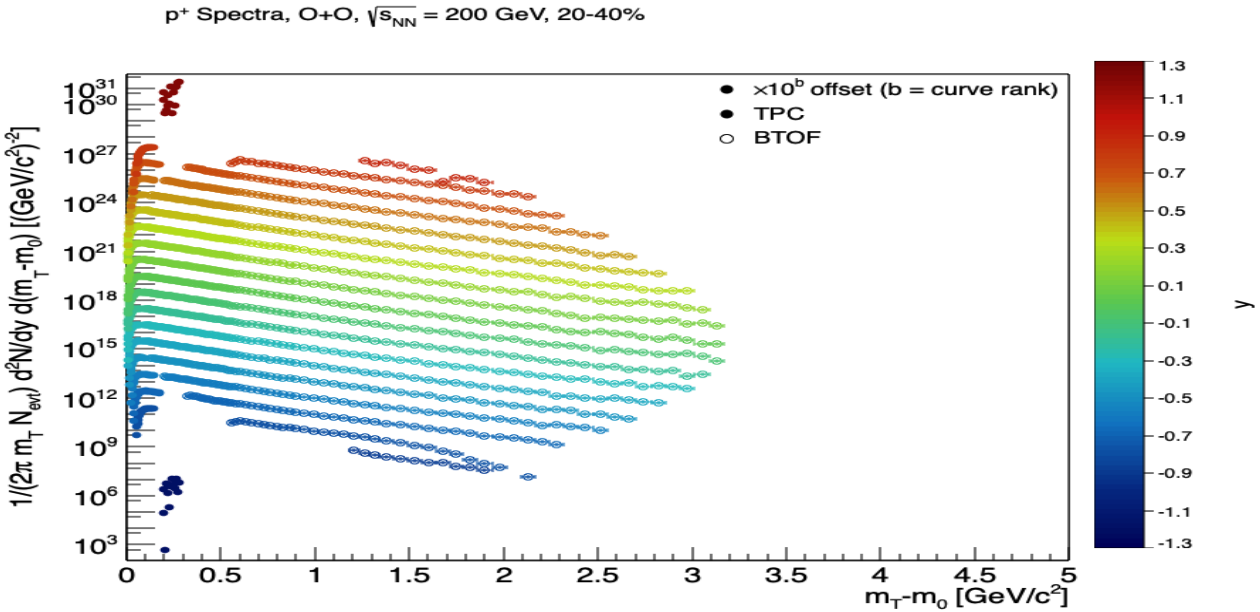
BTOF-corrected proton+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Proton+: 10-20% Centrality



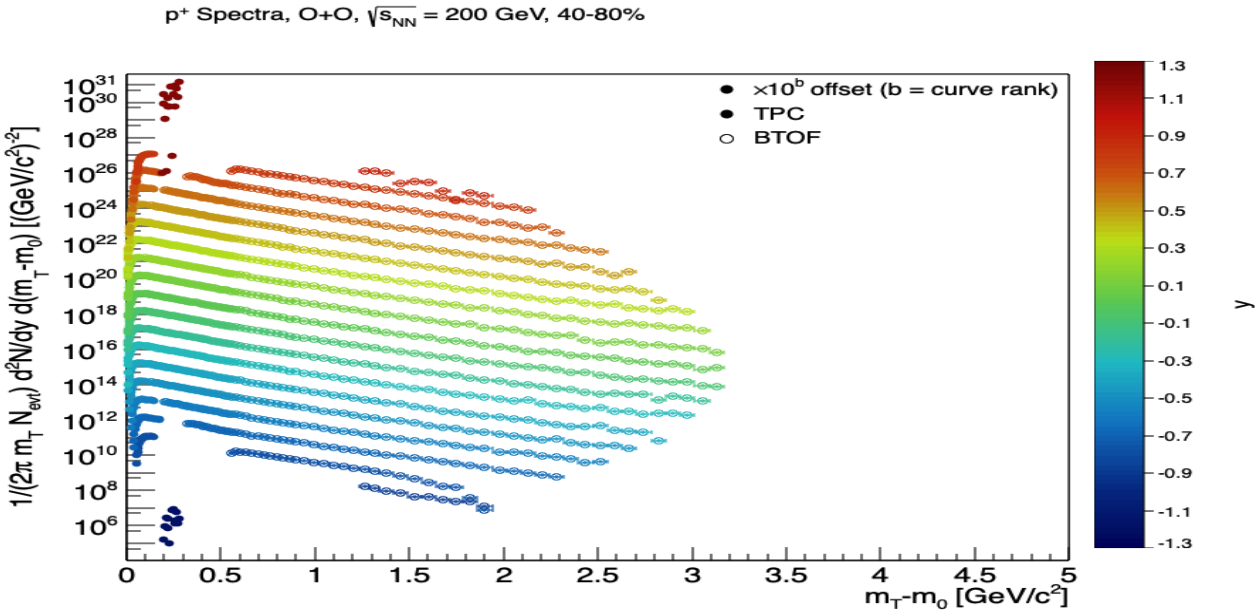
BTOF-corrected proton+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Proton+: 20-40% Centrality



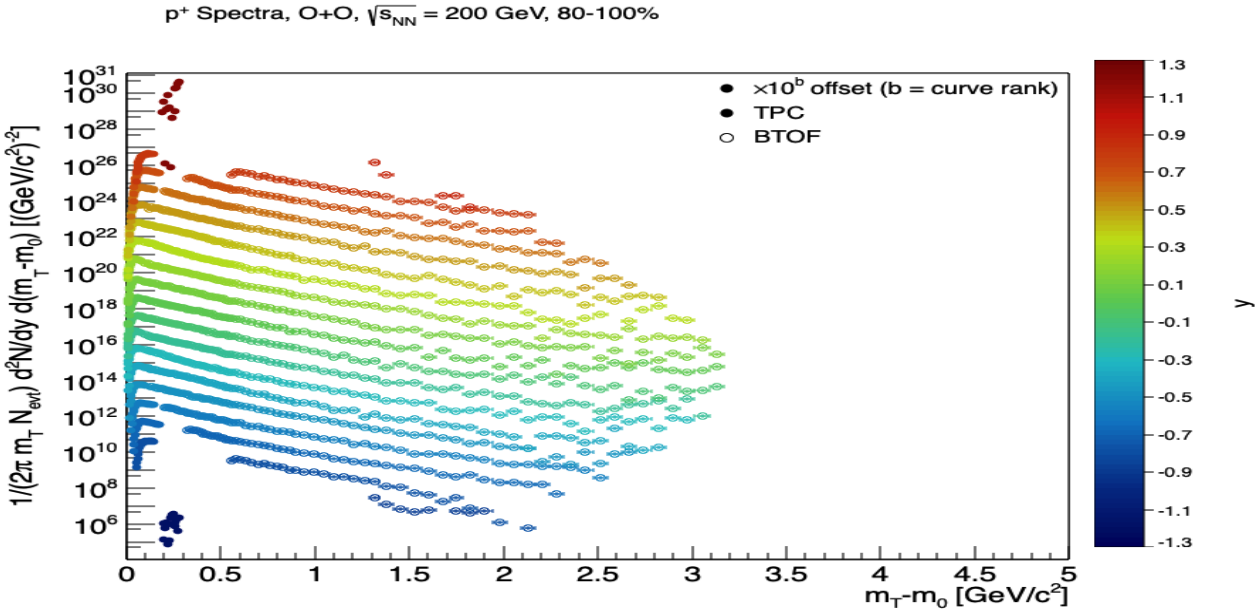
BTOF-corrected proton+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Proton+: 40-80% Centrality



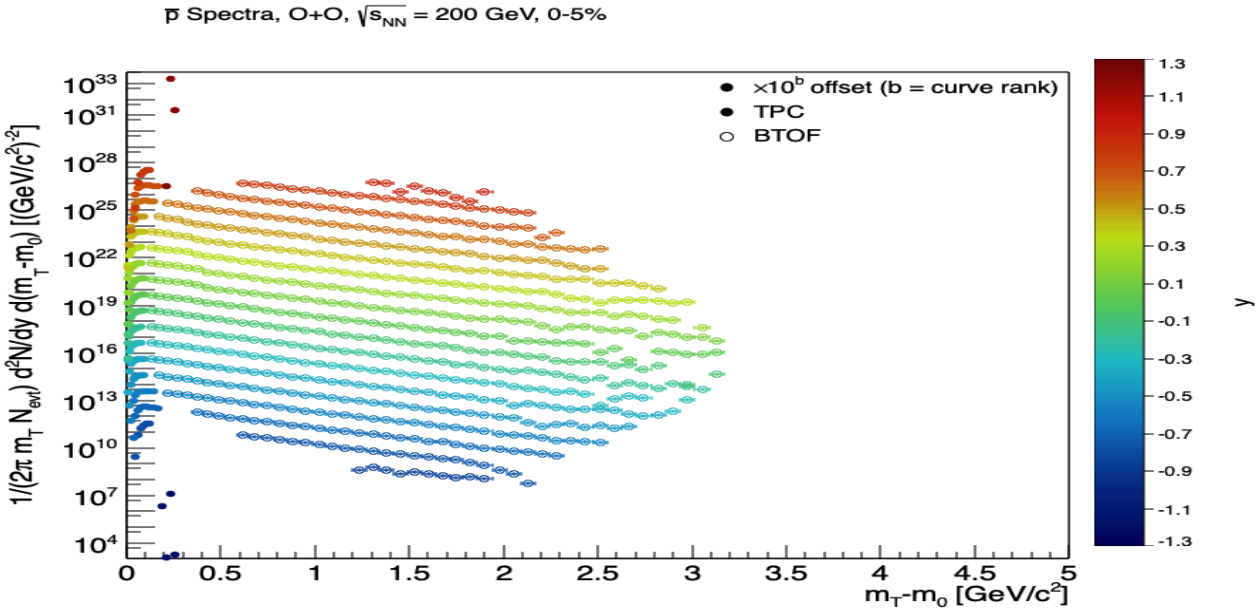
BTOF-corrected proton+ mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Proton+: 80-100% Centrality



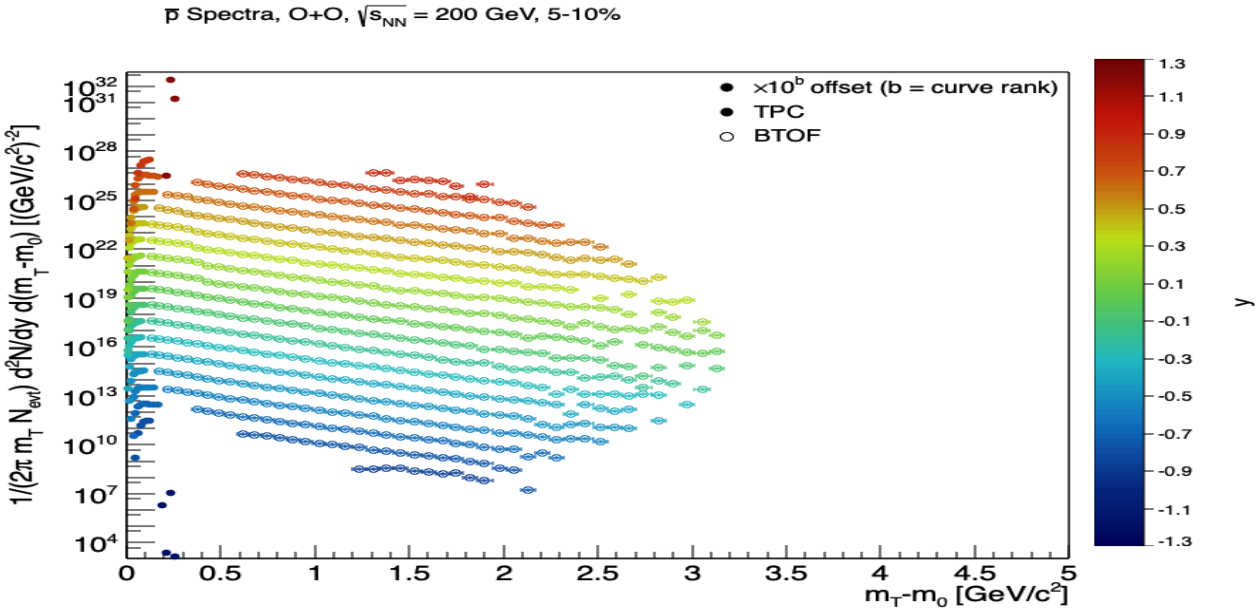
BTOF-corrected proton+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 0-5% Centrality



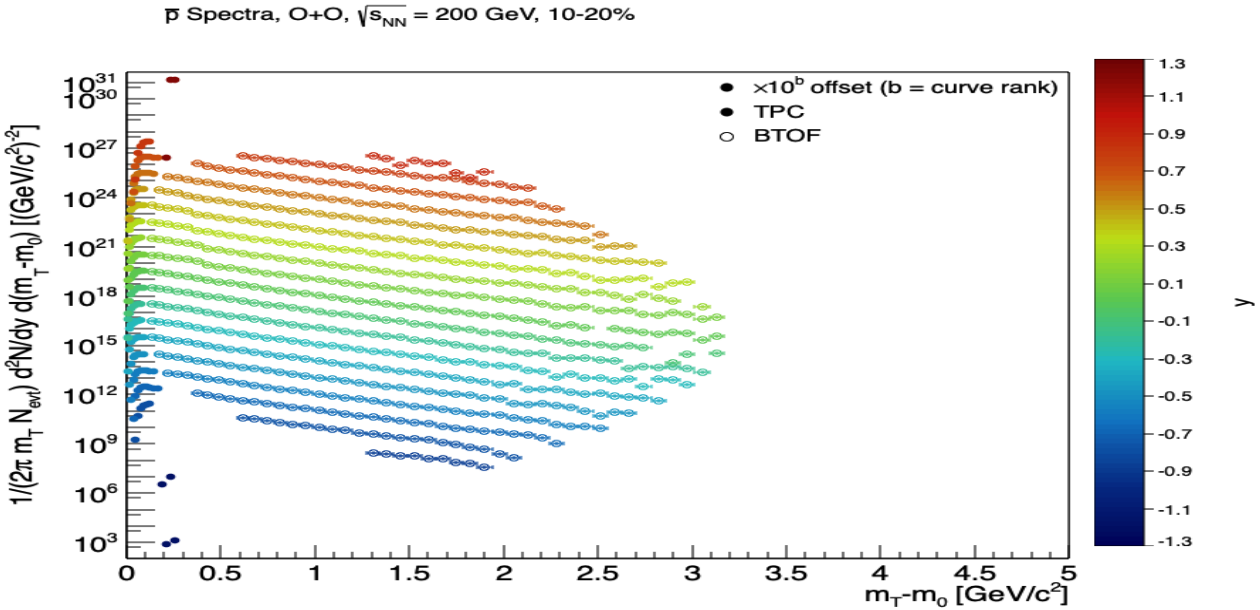
BTOF-corrected antiproton mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 5-10% Centrality



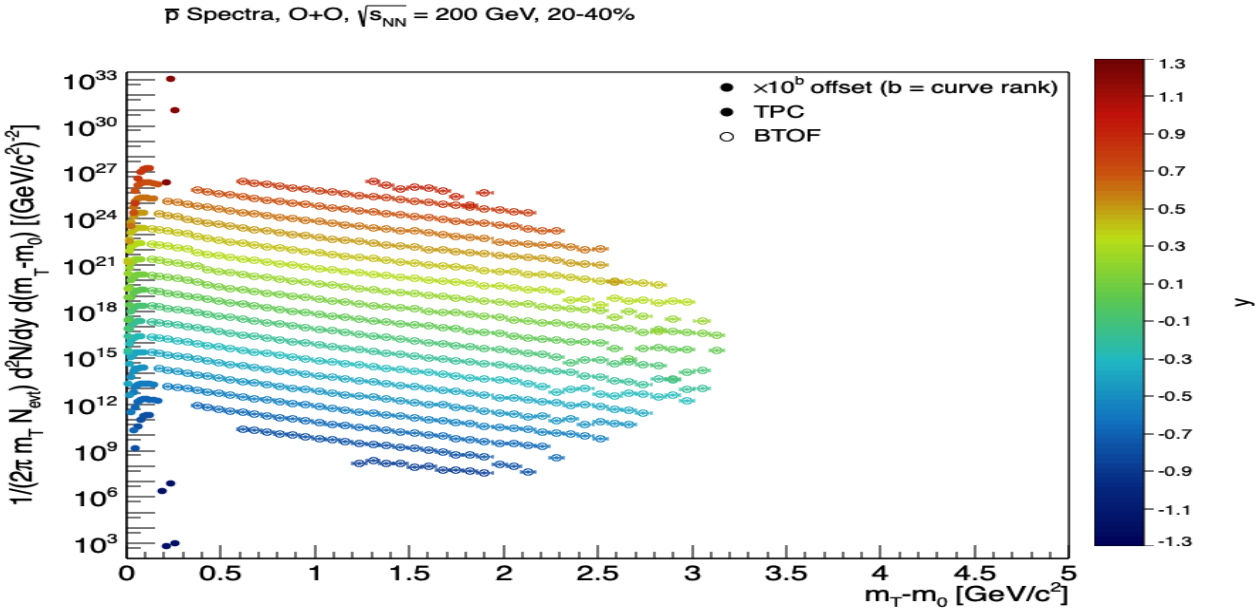
BTOF-corrected antiproton mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 10-20% Centrality



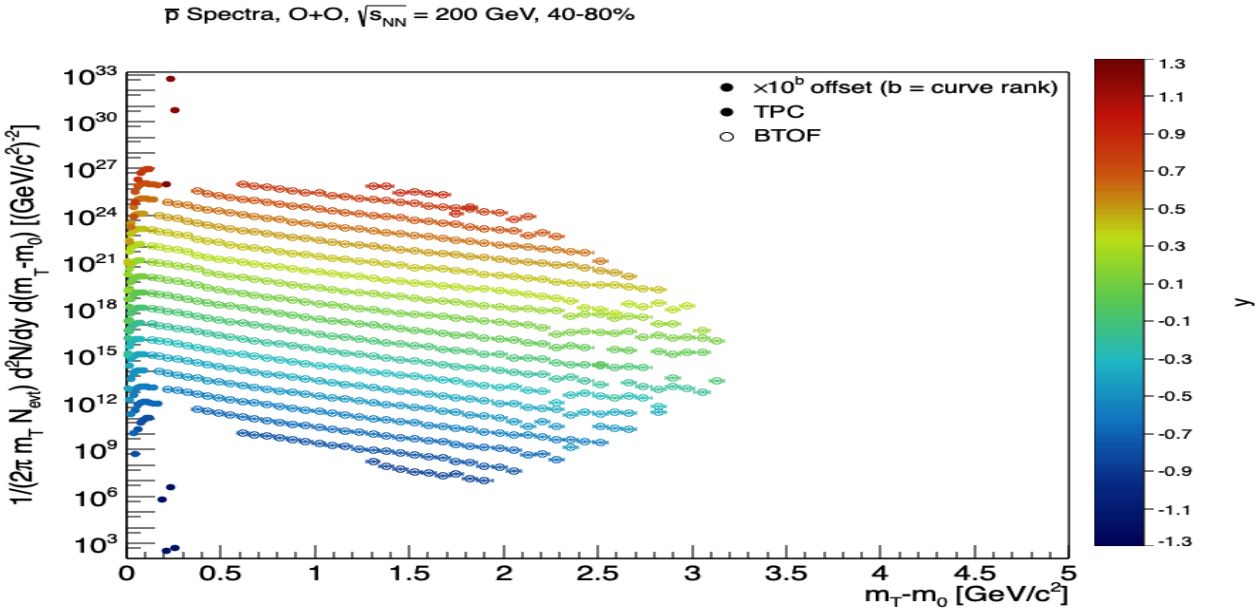
BTOF-corrected antiproton mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 20-40% Centrality



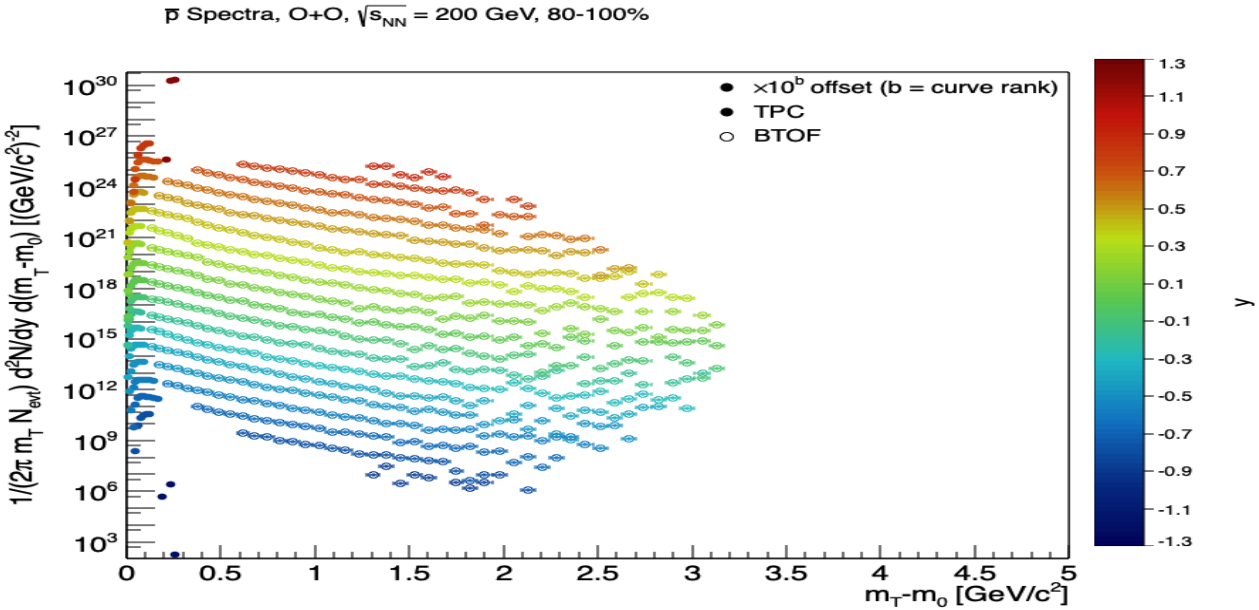
BTOF-corrected antiproton mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 40-80% Centrality



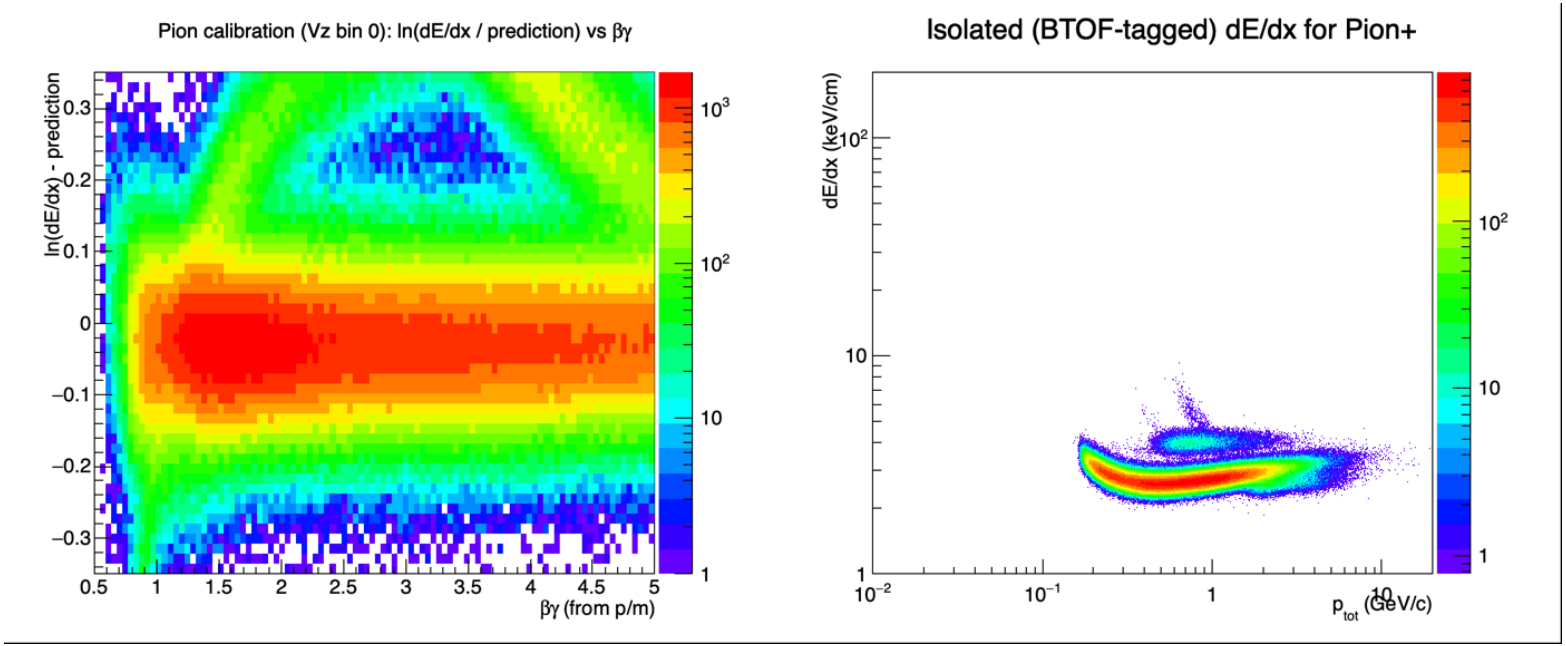
BTOF-corrected antiproton mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 80-100% Centrality



BTOF-corrected antiproton mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

dE/dx Calibration QA



Pion dE/dx calibration residual vs beta-gamma (left) and isolated BTOF-tagged pion+ dE/dx (right) -- calibration is flat and centered.